

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12388086
Kind Code	B2
Date of Patent	August 12, 2025
Inventor(s)	Siu; Carrie et al.

ϵ -VOPO.SUB.4 .cathode for lithium ion batteries

Abstract

The epsilon polymorph of vanadyl phosphate, ϵ -VOPO.sub.4, made from the solvothermally synthesized H.sub.2VOPO.sub.4, is a high-density cathode material for lithium-ion batteries optimized to reversibly intercalate two Li-ions to reach the full theoretical capacity at least 50 cycles with a coulombic efficiency of 98%. This material adopts a stable 3D tunnel structure and can extract two Li-ions per vanadium ion, giving a theoretical capacity of 305 mAh/g, with an upper charge/discharge plateau at around 4.0 V, and one lower at around 2.5 V.

Inventors: Siu; Carrie (Philadelphia, PA), Whittingham; M. Stanley (Vestal, NY)

Applicant: The Research Foundation for The State University of New York (Binghamton, NY)

Family ID: 67767771

Assignee: The Research Foundation for The State University of New York (Binghamton, NY)

Appl. No.: 18/731203

Filed: May 31, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240339610 A1	Oct. 10, 2024

Related U.S. Application Data

division parent-doc US 18354493 20230718 US 12002957 child-doc US 18731203
division parent-doc US 17670470 20220213 US 11715829 child-doc US 18354493
division parent-doc US 16291617 20190304 US 11251430 20220215 child-doc US 17670470
us-provisional-application US 62638893 20180305

Publication Classification

Int. Cl.: **H01M4/58** (20100101); **C01B25/37** (20060101); **H01M4/02** (20060101); **H01M4/136** (20100101); **H01M4/1397** (20100101); **H01M4/62** (20060101); **H01M10/0525** (20100101)

U.S. Cl.:

CPC **H01M4/5825** (20130101); **C01B25/372** (20130101); **H01M4/136** (20130101); **H01M4/1397** (20130101); **H01M4/623** (20130101); **H01M4/625** (20130101); **H01M10/0525** (20130101); C01P2002/72 (20130101); C01P2002/90 (20130101); C01P2004/03 (20130101); C01P2004/04 (20130101); H01M2004/021 (20130101); H01M2004/028 (20130101)

Field of Classification Search

CPC: H01M (4/136); H01M (4/5825); H01M (4/623); H01M (4/625); H01M (2004/028); C01B (25/372)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4355162	12/1981	Johnson	N/A	N/A
4376709	12/1982	Johnson	N/A	N/A
4459406	12/1983	Johnson	N/A	N/A
4465743	12/1983	Skarstad	N/A	N/A
4619874	12/1985	Skarstad et al.	N/A	N/A
4744787	12/1987	Phipps et al.	N/A	N/A
5061581	12/1990	Narang et al.	N/A	N/A
5114809	12/1991	Nakacho et al.	N/A	N/A
5135477	12/1991	Untereker et al.	N/A	N/A
5153080	12/1991	Inubushi et al.	N/A	N/A
5415628	12/1994	Untereker et al.	N/A	N/A
5419890	12/1994	Saidi	N/A	N/A
5437692	12/1994	Dasgupta et al.	N/A	N/A
5443809	12/1994	Olsen	N/A	N/A
5453261	12/1994	Saidi et al.	N/A	N/A
5482697	12/1995	Saidi	N/A	N/A
5498489	12/1995	Dasgupta et al.	N/A	N/A
5512389	12/1995	Dasgupta et al.	N/A	N/A
5573503	12/1995	Untereker et al.	N/A	N/A
5576120	12/1995	Saidi	N/A	N/A
5599643	12/1996	Plichta et al.	N/A	N/A
5639577	12/1996	Takeuchi et al.	N/A	N/A
5667916	12/1996	Ebel et al.	N/A	N/A
5711761	12/1997	Untereker et al.	N/A	N/A
5731104	12/1997	Ventura et al.	N/A	N/A
5759715	12/1997	Barker et al.	N/A	N/A

5849434	12/1997	Miura et al.	N/A	N/A
5851696	12/1997	Saidi et al.	N/A	N/A
6007945	12/1998	Jacobs et al.	N/A	N/A
6015638	12/1999	Ventura et al.	N/A	N/A
6124057	12/1999	Ito et al.	N/A	N/A
6156395	12/1999	Zhang et al.	N/A	N/A
6225007	12/2000	Horne et al.	N/A	N/A
6331282	12/2000	Manthiram et al.	N/A	N/A
6337155	12/2001	Takei et al.	N/A	N/A
6361755	12/2001	Manthiram et al.	N/A	N/A
6391494	12/2001	Reitz et al.	N/A	N/A
6403253	12/2001	Wainwright et al.	N/A	N/A
6492061	12/2001	Gauthier et al.	N/A	N/A
6503646	12/2002	Ghantous et al.	N/A	N/A
6528033	12/2002	Barker et al.	N/A	N/A
6645452	12/2002	Barker et al.	N/A	N/A
6653022	12/2002	Nordlinder et al.	N/A	N/A
6702961	12/2003	Barker et al.	N/A	N/A
6716372	12/2003	Barker et al.	N/A	N/A
6723470	12/2003	Barker et al.	N/A	N/A
6730437	12/2003	Leising et al.	N/A	N/A
6749966	12/2003	Reitz et al.	N/A	N/A
6805999	12/2003	Lee et al.	N/A	N/A
6838023	12/2004	Torardi	N/A	N/A
6844047	12/2004	Kaminsky et al.	N/A	N/A
6855273	12/2004	Ravet et al.	N/A	N/A
6859297	12/2004	Lee et al.	N/A	N/A
6872492	12/2004	Barker et al.	N/A	N/A
6884544	12/2004	Barker et al.	N/A	N/A
6888663	12/2004	Bourdelaïs et al.	N/A	N/A
6913855	12/2004	Stoker et al.	N/A	N/A
6958860	12/2004	Dontula et al.	N/A	N/A
6960331	12/2004	Barker et al.	N/A	N/A
6962666	12/2004	Ravet et al.	N/A	N/A
7001690	12/2005	Barker et al.	N/A	N/A
7026072	12/2005	Barker et al.	N/A	N/A
7041239	12/2005	Barker et al.	N/A	N/A
7045249	12/2005	Nanjundaswamy et al.	N/A	N/A
7192564	12/2006	Cardarelli et al.	N/A	N/A
7198869	12/2006	Ghantous et al.	N/A	N/A
7300722	12/2006	Christian et al.	N/A	N/A
7338734	12/2007	Chiang et al.	N/A	N/A
7344659	12/2007	Ravet et al.	N/A	N/A
7348100	12/2007	Adamson et al.	N/A	N/A
7383083	12/2007	Fischer et al.	N/A	N/A
7422823	12/2007	Saidi et al.	N/A	N/A
7438992	12/2007	Barker et al.	N/A	N/A
7482097	12/2008	Saidi et al.	N/A	N/A
7632602	12/2008	Lee et al.	N/A	N/A
7722787	12/2009	Horne et al.	N/A	N/A

7759008	12/2009	Barker et al.	N/A	N/A
7771628	12/2009	Barker et al.	N/A	N/A
7790319	12/2009	Faulkner	N/A	N/A
7815819	12/2009	Ravet et al.	N/A	N/A
7901810	12/2010	Barker et al.	N/A	N/A
7906239	12/2010	Watanabe et al.	N/A	N/A
7927742	12/2010	Scott et al.	N/A	N/A
7939201	12/2010	Chiang et al.	N/A	N/A
7952090	12/2010	Kugler	N/A	N/A
7955733	12/2010	Armand et al.	N/A	N/A
7960058	12/2010	Goodenough et al.	N/A	N/A
7960059	12/2010	Adamson et al.	N/A	N/A
7964308	12/2010	Goodenough et al.	N/A	N/A
7972728	12/2010	Goodenough et al.	N/A	N/A
7988746	12/2010	Chiang et al.	N/A	N/A
7998617	12/2010	Armand et al.	N/A	N/A
8003250	12/2010	Chiang et al.	N/A	N/A
8057936	12/2010	Chiang et al.	N/A	N/A
8067117	12/2010	Goodenough et al.	N/A	N/A
8088512	12/2011	Gozdz et al.	N/A	N/A
8119285	12/2011	Hatta et al.	N/A	N/A
8148009	12/2011	Chiang et al.	N/A	N/A
8148013	12/2011	Chiang et al.	N/A	N/A
8158090	12/2011	Chiang et al.	N/A	N/A
8168326	12/2011	Chiang et al.	N/A	N/A
8173049	12/2011	Ravet et al.	N/A	N/A
8178239	12/2011	Hatta et al.	N/A	N/A
8206468	12/2011	Chiang et al.	N/A	N/A
8206469	12/2011	Chiang et al.	N/A	N/A
8218351	12/2011	Bloch et al.	N/A	N/A
8241789	12/2011	Chiang et al.	N/A	N/A
8241792	12/2011	Kaneko et al.	N/A	N/A
8257616	12/2011	Ravet et al.	N/A	N/A
8277975	12/2011	Chiang et al.	N/A	N/A
8282691	12/2011	Goodenough et al.	N/A	N/A
8309242	12/2011	Wei et al.	N/A	N/A
8318352	12/2011	Saidi et al.	N/A	N/A
8318353	12/2011	Faulkner	N/A	N/A
8323831	12/2011	Wilkins et al.	N/A	N/A
8323832	12/2011	Chiang et al.	N/A	N/A
8329071	12/2011	Wang et al.	N/A	N/A
8337723	12/2011	Hashiba et al.	N/A	N/A
8361661	12/2012	Doe et al.	N/A	N/A
8383269	12/2012	Scott et al.	N/A	N/A
8420258	12/2012	Rojeski	N/A	N/A
8435678	12/2012	Wixom et al.	N/A	N/A
8435738	12/2012	Holmes	N/A	N/A
8445135	12/2012	Sano et al.	N/A	N/A
8449980	12/2012	Suzuki et al.	N/A	N/A
8475739	12/2012	Holmes et al.	N/A	N/A

8481208	12/2012	Riley, Jr. et al.	N/A	N/A
8491861	12/2012	Evenson et al.	N/A	N/A
8492031	12/2012	Suzuki et al.	N/A	N/A
8506847	12/2012	Hibst et al.	N/A	N/A
8506851	12/2012	Ravet et al.	N/A	N/A
8506852	12/2012	Ravet et al.	N/A	N/A
8580430	12/2012	Chiang et al.	N/A	N/A
8586238	12/2012	Chiang et al.	N/A	N/A
8614020	12/2012	Kondo et al.	N/A	N/A
8617430	12/2012	Chiang et al.	N/A	N/A
8617745	12/2012	Gozdz et al.	N/A	N/A
8632904	12/2013	Guyomard et al.	N/A	N/A
8652683	12/2013	Rojeski	N/A	N/A
8658310	12/2013	Rojeski	N/A	N/A
8709647	12/2013	Gozdz et al.	N/A	N/A
8722227	12/2013	Chiang et al.	N/A	N/A
8722242	12/2013	Muldoon et al.	N/A	N/A
8734539	12/2013	Tokita et al.	N/A	N/A
8784694	12/2013	Kay	N/A	N/A
8785043	12/2013	Armand et al.	N/A	N/A
8798235	12/2013	Ohta et al.	N/A	N/A
8798236	12/2013	Ohta et al.	N/A	N/A
8802295	12/2013	Nesper et al.	N/A	N/A
8808575	12/2013	Hibst et al.	N/A	N/A
8808576	12/2013	Hibst et al.	N/A	N/A
8821763	12/2013	Sano et al.	N/A	N/A
8822080	12/2013	Fujino et al.	N/A	N/A
8840838	12/2013	Holmes et al.	N/A	N/A
8852807	12/2013	Chiang et al.	N/A	N/A
8870976	12/2013	Kondo et al.	N/A	N/A
8895193	12/2013	Abouimrane et al.	N/A	N/A
8926389	12/2014	Nishido	N/A	N/A
8929510	12/2014	Nishino et al.	N/A	N/A
8932762	12/2014	Sano et al.	N/A	N/A
8936871	12/2014	Sano et al.	N/A	N/A
8951676	12/2014	Doe et al.	N/A	N/A
8951680	12/2014	Doe et al.	N/A	N/A
8957442	12/2014	Seo et al.	N/A	N/A
8980453	12/2014	Scott et al.	N/A	N/A
8993171	12/2014	Sano et al.	N/A	N/A
8999282	12/2014	Ceder et al.	N/A	N/A
8999571	12/2014	Chiang et al.	N/A	N/A
9000458	12/2014	Seo et al.	N/A	N/A
9006755	12/2014	Seo et al.	N/A	N/A
9029003	12/2014	Sulaiman Lo et al.	N/A	N/A
9044191	12/2014	Nishino et al.	N/A	N/A
9065093	12/2014	Chiang et al.	N/A	N/A
9077032	12/2014	Doe et al.	N/A	N/A
9077037	12/2014	Hwu et al.	N/A	N/A
9093707	12/2014	Lee et al.	N/A	N/A

9105869	12/2014	Nishido	N/A	N/A
9112210	12/2014	Chen et al.	N/A	N/A
9128015	12/2014	Holmes et al.	N/A	N/A
9136505	12/2014	Sasaki et al.	N/A	N/A
9147874	12/2014	Chen et al.	N/A	N/A
9153839	12/2014	Furuya	N/A	N/A
9168016	12/2014	Ohta et al.	N/A	N/A
9172111	12/2014	Doe et al.	N/A	N/A
9190659	12/2014	Kondo et al.	N/A	N/A
9196923	12/2014	Chang et al.	N/A	N/A
9203084	12/2014	Wang et al.	N/A	N/A
9240612	12/2015	Doe et al.	N/A	N/A
9250229	12/2015	Holmes	N/A	N/A
9257655	12/2015	Suzuki et al.	N/A	N/A
9257718	12/2015	Kondo et al.	N/A	N/A
9258853	12/2015	Nishido et al.	N/A	N/A
9268915	12/2015	Holmes et al.	N/A	N/A
9287578	12/2015	Anderson et al.	N/A	N/A
9287580	12/2015	Jain	N/A	N/A
9293790	12/2015	Doe et al.	N/A	N/A
9296621	12/2015	Feng et al.	N/A	N/A
9299966	12/2015	Chiang et al.	N/A	N/A
9305716	12/2015	Tan et al.	N/A	N/A
9318739	12/2015	Toya et al.	N/A	N/A
9318742	12/2015	Fujino et al.	N/A	N/A
9331283	12/2015	Lim et al.	N/A	N/A
9337481	12/2015	Nesper et al.	N/A	N/A
9349544	12/2015	Rojeski	N/A	N/A
9350018	12/2015	Tsutsumi et al.	N/A	N/A
9362555	12/2015	Chen et al.	N/A	N/A
9362562	12/2015	Armand et al.	N/A	N/A
9368831	12/2015	He et al.	N/A	N/A
9385364	12/2015	Liang et al.	N/A	N/A
9391290	12/2015	Watabe et al.	N/A	N/A
9401528	12/2015	Doe et al.	N/A	N/A
9406930	12/2015	Fukui et al.	N/A	N/A
9413006	12/2015	Kay	N/A	N/A
9419279	12/2015	Yamamura et al.	N/A	N/A
9437866	12/2015	Kikuchi et al.	N/A	N/A
9449765	12/2015	Gardner et al.	N/A	N/A
9450209	12/2015	Seo et al.	N/A	N/A
9461092	12/2015	Seo et al.	N/A	N/A
9472601	12/2015	Seo et al.	N/A	N/A
9515310	12/2015	Cao et al.	N/A	N/A
9525164	12/2015	Aksu et al.	N/A	N/A
9527157	12/2015	Kroll et al.	N/A	N/A
9537109	12/2016	Suzuki et al.	N/A	N/A
9548509	12/2016	Anderson et al.	N/A	N/A
9553312	12/2016	Yamauchi et al.	N/A	N/A
9559351	12/2016	Mori et al.	N/A	N/A

9564641	12/2016	Otsuki et al.	N/A	N/A
9581877	12/2016	Bass et al.	N/A	N/A
9583764	12/2016	Yamauchi et al.	N/A	N/A
9592508	12/2016	Holmes et al.	N/A	N/A
9595697	12/2016	Nishido et al.	N/A	N/A
9608269	12/2016	Barker	N/A	N/A
9627691	12/2016	Xing et al.	N/A	N/A
9632102	12/2016	Holmes et al.	N/A	N/A
9634267	12/2016	Watabe et al.	N/A	N/A
9645143	12/2016	Holmes et al.	N/A	N/A
9653517	12/2016	Uesaka et al.	N/A	N/A
9653730	12/2016	Pitteloud	N/A	N/A
9664702	12/2016	Holmes et al.	N/A	N/A
9666755	12/2016	Shimomura	N/A	N/A
9683314	12/2016	Xu et al.	N/A	N/A
9685656	12/2016	Yamauchi et al.	N/A	N/A
9711793	12/2016	Mizuno et al.	N/A	N/A
9719990	12/2016	Holmes et al.	N/A	N/A
9722247	12/2016	Zhang et al.	N/A	N/A
9745194	12/2016	Kabe et al.	N/A	N/A
9771319	12/2016	Gonzalez Martinez et al.	N/A	N/A
9786913	12/2016	Inoue et al.	N/A	N/A
9786944	12/2016	Chiang et al.	N/A	N/A
9793061	12/2016	Gardner et al.	N/A	N/A
9798214	12/2016	Garcia et al.	N/A	N/A
9810704	12/2016	Holmes et al.	N/A	N/A
9819054	12/2016	Kang et al.	N/A	N/A
9853242	12/2016	Nishido	N/A	N/A
9859557	12/2017	Mori et al.	N/A	N/A
9865877	12/2017	Kang et al.	N/A	N/A
9878305	12/2017	Hossain et al.	N/A	N/A
9882204	12/2017	Ryoshi et al.	N/A	N/A
9887047	12/2017	Tan et al.	N/A	N/A
9899672	12/2017	Zhamu et al.	N/A	N/A
9899710	12/2017	Scott et al.	N/A	N/A
9911985	12/2017	Dong et al.	N/A	N/A
9917271	12/2017	Uesaka et al.	N/A	N/A
9923206	12/2017	Chen et al.	N/A	N/A
9929368	12/2017	Kataishi et al.	N/A	N/A
9935158	12/2017	Seo et al.	N/A	N/A
9935286	12/2017	Watabe et al.	N/A	N/A
9935314	12/2017	Johnson et al.	N/A	N/A
9942972	12/2017	Nishino et al.	N/A	N/A
9952240	12/2017	Holmes et al.	N/A	N/A
9954224	12/2017	Toya et al.	N/A	N/A
9960451	12/2017	Zhamu et al.	N/A	N/A
9966604	12/2017	Kondo et al.	N/A	N/A
9997778	12/2017	Cao et al.	N/A	N/A
10007163	12/2017	Bass et al.	N/A	N/A

10008723	12/2017	Zhamu et al.	N/A	N/A
10008747	12/2017	Zhamu et al.	N/A	N/A
10012664	12/2017	Wasson et al.	N/A	N/A
10017875	12/2017	Toya et al.	N/A	N/A
10018643	12/2017	Holmes et al.	N/A	N/A
10044025	12/2017	Toya et al.	N/A	N/A
10050271	12/2017	Barker et al.	N/A	N/A
10069097	12/2017	Kataishi et al.	N/A	N/A
10069138	12/2017	Mizuno et al.	N/A	N/A
10084156	12/2017	Seo et al.	N/A	N/A
10090516	12/2017	Balaya et al.	N/A	N/A
10096658	12/2017	Watabe et al.	N/A	N/A
10109849	12/2017	Ryoshi et al.	N/A	N/A
10109859	12/2017	Goodenough et al.	N/A	N/A
10128501	12/2017	Fukui et al.	N/A	N/A
10130936	12/2017	Hossain et al.	N/A	N/A
10134998	12/2017	Seo et al.	N/A	N/A
10153516	12/2017	Schmitz et al.	N/A	N/A
10160660	12/2017	Rodriguez et al.	N/A	N/A
10170212	12/2018	Barker et al.	N/A	N/A
10170749	12/2018	Zhamu et al.	N/A	N/A
10170789	12/2018	Zhamu et al.	N/A	N/A
10199637	12/2018	Zhamu et al.	N/A	N/A
10199647	12/2018	Chen et al.	N/A	N/A
10201065	12/2018	Ohta et al.	N/A	N/A
10207992	12/2018	Suzuki et al.	N/A	N/A
10224494	12/2018	Watabe et al.	N/A	N/A
10230055	12/2018	Kanamoto et al.	N/A	N/A
10230108	12/2018	Sano et al.	N/A	N/A
10243217	12/2018	Zhamu et al.	N/A	N/A
10263253	12/2018	Nair et al.	N/A	N/A
10263277	12/2018	Pitts et al.	N/A	N/A
10263308	12/2018	McKone et al.	N/A	N/A
10270098	12/2018	Kang et al.	N/A	N/A
10276856	12/2018	Zhamu et al.	N/A	N/A
10333138	12/2018	Goodenough et al.	N/A	N/A
10340470	12/2018	Sasaki et al.	N/A	N/A
10361388	12/2018	Mitsumori et al.	N/A	N/A
10361460	12/2018	Zhamu et al.	N/A	N/A
10371710	12/2018	Wasson et al.	N/A	N/A
10388983	12/2018	Zhamu et al.	N/A	N/A
10396356	12/2018	Toya et al.	N/A	N/A
10411264	12/2018	Pan et al.	N/A	N/A
10418662	12/2018	Zhamu et al.	N/A	N/A
10424787	12/2018	Kamata et al.	N/A	N/A
10446845	12/2018	Goodenough et al.	N/A	N/A
10446853	12/2018	Choi et al.	N/A	N/A
10461134	12/2018	Seo et al.	N/A	N/A
10483533	12/2018	Zhamu et al.	N/A	N/A
10490815	12/2018	Toya et al.	N/A	N/A

10493432	12/2018	Islam et al.	N/A	N/A
10516165	12/2018	Toya et al.	N/A	N/A
10518265	12/2018	Holmes et al.	N/A	N/A
10534009	12/2019	Holmes et al.	N/A	N/A
10535880	12/2019	Zhamu et al.	N/A	N/A
10535892	12/2019	Zhamu et al.	N/A	N/A
10553857	12/2019	Mizuno et al.	N/A	N/A
10557863	12/2019	Holmes et al.	N/A	N/A
10559815	12/2019	Zhamu et al.	N/A	N/A
10573932	12/2019	Yang et al.	N/A	N/A
10586931	12/2019	Kanamoto et al.	N/A	N/A
10611639	12/2019	Sano et al.	N/A	N/A
10615408	12/2019	Fellner et al.	N/A	N/A
10615453	12/2019	Zanotti et al.	N/A	N/A
10615463	12/2019	Scott et al.	N/A	N/A
10627418	12/2019	Wasson et al.	N/A	N/A
10637043	12/2019	Zhamu et al.	N/A	N/A
10637115	12/2019	Licht	N/A	N/A
10644250	12/2019	Watabe et al.	N/A	N/A
10644308	12/2019	Liu et al.	N/A	N/A
10658604	12/2019	Mitsumori et al.	N/A	N/A
10661090	12/2019	Chen et al.	N/A	N/A
10669646	12/2019	Toya et al.	N/A	N/A
10673090	12/2019	Nie et al.	N/A	N/A
10693094	12/2019	Seo et al.	N/A	N/A
10697082	12/2019	Huang et al.	N/A	N/A
10698286	12/2019	Bass et al.	N/A	N/A
10727479	12/2019	Zhamu et al.	N/A	N/A
10727531	12/2019	Zhamu et al.	N/A	N/A
10734635	12/2019	Zhamu et al.	N/A	N/A
10734646	12/2019	He et al.	N/A	N/A
10734648	12/2019	Zhamu et al.	N/A	N/A
10734671	12/2019	Zhamu et al.	N/A	N/A
10741619	12/2019	Seo et al.	N/A	N/A
10741832	12/2019	Sun et al.	N/A	N/A
10749168	12/2019	Johnson et al.	N/A	N/A
10756286	12/2019	Ishisone et al.	N/A	N/A
10763491	12/2019	Takeuchi et al.	N/A	N/A
10766795	12/2019	Su et al.	N/A	N/A
10770721	12/2019	Zhamu et al.	N/A	N/A
10772199	12/2019	Lai et al.	N/A	N/A
10777810	12/2019	He et al.	N/A	N/A
10784509	12/2019	He et al.	N/A	N/A
10790509	12/2019	Uekusa et al.	N/A	N/A
10797113	12/2019	Ohsawa et al.	N/A	N/A
10797313	12/2019	Zhamu et al.	N/A	N/A
10818921	12/2019	Kensaku et al.	N/A	N/A
10826132	12/2019	Smith et al.	N/A	N/A
10829385	12/2019	Matsumoto et al.	N/A	N/A
10840501	12/2019	Hong et al.	N/A	N/A

10840505	12/2019	Zarra et al.	N/A	N/A
10840565	12/2019	Zhamu et al.	N/A	N/A
10854396	12/2019	Yokota et al.	N/A	N/A
10862129	12/2019	Pan et al.	N/A	N/A
10868256	12/2019	Seo et al.	N/A	N/A
10873083	12/2019	Zhamu et al.	N/A	N/A
10886525	12/2020	Singh et al.	N/A	N/A
10886528	12/2020	Pan et al.	N/A	N/A
10889506	12/2020	Rodriguez et al.	N/A	N/A
10903519	12/2020	Pan et al.	N/A	N/A
10903527	12/2020	Liu et al.	N/A	N/A
10910568	12/2020	Watabe et al.	N/A	N/A
10910576	12/2020	Mitsumori et al.	N/A	N/A
10916769	12/2020	Su et al.	N/A	N/A
10923717	12/2020	Kim et al.	N/A	N/A
10930924	12/2020	Zhamu et al.	N/A	N/A
10937965	12/2020	Seo et al.	N/A	N/A
2001/0046468	12/2000	Reitz et al.	N/A	N/A
2002/0039687	12/2001	Barker et al.	N/A	N/A
2002/0064497	12/2001	Horne et al.	N/A	N/A
2002/0086212	12/2001	Nordlinder et al.	N/A	N/A
2002/0086214	12/2001	Barker et al.	N/A	N/A
2002/0142218	12/2001	Reitz et al.	N/A	N/A
2002/0192553	12/2001	Barker et al.	N/A	N/A
2002/0195591	12/2001	Ravet et al.	N/A	N/A
2003/0003362	12/2002	Leising et al.	N/A	N/A
2003/0022065	12/2002	Zhang et al.	N/A	N/A
2003/0073003	12/2002	Barker et al.	N/A	N/A
2003/0077513	12/2002	Ghantous et al.	N/A	N/A
2003/0077514	12/2002	Barker et al.	N/A	N/A
2003/0129492	12/2002	Barker et al.	N/A	N/A
2003/0162094	12/2002	Lee et al.	N/A	N/A
2003/0190526	12/2002	Saidi et al.	N/A	N/A
2003/0190527	12/2002	Pugh et al.	N/A	N/A
2003/0190528	12/2002	Saidi et al.	N/A	N/A
2003/0215714	12/2002	Barker et al.	N/A	N/A
2003/0235761	12/2002	Prabaharan	N/A	N/A
2004/0005265	12/2003	Chiang et al.	N/A	N/A
2004/0005500	12/2003	Torardi	N/A	N/A
2004/0013943	12/2003	Stoker et al.	N/A	N/A
2004/0016632	12/2003	Barker	N/A	N/A
2004/0028585	12/2003	Cardarelli et al.	N/A	N/A
2004/0048157	12/2003	Neudecker et al.	N/A	N/A
2004/0066556	12/2003	Dontula et al.	N/A	N/A
2004/0121195	12/2003	Ghantous et al.	N/A	N/A
2004/0126300	12/2003	Barker et al.	N/A	N/A
2004/0140458	12/2003	Ravet et al.	N/A	N/A
2004/0150867	12/2003	Lee et al.	N/A	N/A
2004/0197654	12/2003	Barker et al.	N/A	N/A
2004/0262571	12/2003	Barker et al.	N/A	N/A

2005/0003269	12/2004	Nanjundaswamy et al.	N/A	N/A
2005/0181283	12/2004	Pugh et al.	N/A	N/A
2005/0186476	12/2004	Barker et al.	N/A	N/A
2005/0194567	12/2004	Adamson et al.	N/A	N/A
2005/0238961	12/2004	Barker et al.	N/A	N/A
2006/0060827	12/2005	Ravet et al.	N/A	N/A
2006/0088767	12/2005	Li et al.	N/A	N/A
2006/0110580	12/2005	Aylward et al.	N/A	N/A
2006/0194112	12/2005	Barker et al.	N/A	N/A
2006/0228629	12/2005	Christian et al.	N/A	N/A
2006/0292444	12/2005	Chiang et al.	N/A	N/A
2007/0009802	12/2006	Lee et al.	N/A	N/A
2007/0031732	12/2006	Chiang et al.	N/A	N/A
2007/0031735	12/2006	Nishimura et al.	N/A	N/A
2007/0065724	12/2006	Barker et al.	N/A	N/A
2007/0072034	12/2006	Barker et al.	N/A	N/A
2007/0082262	12/2006	Nishimura et al.	N/A	N/A
2007/0141468	12/2006	Barker	N/A	N/A
2007/0166617	12/2006	Gozdz et al.	N/A	N/A
2007/0190418	12/2006	Chiang et al.	N/A	N/A
2007/0248520	12/2006	Faulkner	N/A	N/A
2007/0292747	12/2006	Chiang et al.	N/A	N/A
2007/0292757	12/2006	Watanabe et al.	N/A	N/A
2007/0298317	12/2006	Brodd et al.	N/A	N/A
2008/0014503	12/2007	Kejha	N/A	N/A
2008/0014507	12/2007	Kejha	N/A	N/A
2008/0052900	12/2007	Kondo et al.	N/A	N/A
2008/0057390	12/2007	Kondo et al.	N/A	N/A
2008/0070113	12/2007	Kondo et al.	N/A	N/A
2008/0131777	12/2007	Hatta et al.	N/A	N/A
2008/0138709	12/2007	Hatta et al.	N/A	N/A
2008/0157024	12/2007	Adamson et al.	N/A	N/A
2008/0257721	12/2007	Ravet et al.	N/A	N/A
2008/0261113	12/2007	Huang et al.	N/A	N/A
2009/0040587	12/2008	Kugler	N/A	N/A
2009/0061314	12/2008	Dong	N/A	N/A
2009/0081553	12/2008	Kondo et al.	N/A	N/A
2009/0214944	12/2008	Rojeski	N/A	N/A
2009/0220838	12/2008	Barker et al.	N/A	N/A
2009/0246636	12/2008	Chiang et al.	N/A	N/A
2009/0253036	12/2008	Owen et al.	N/A	N/A
2009/0272949	12/2008	Buttry	N/A	N/A
2009/0303660	12/2008	Nair et al.	N/A	N/A
2009/0311597	12/2008	Chiang et al.	N/A	N/A
2009/0311680	12/2008	Nakao et al.	N/A	N/A
2010/0055573	12/2009	Lee et al.	N/A	N/A
2010/0075225	12/2009	Wilkins et al.	N/A	N/A
2010/0078591	12/2009	Sano et al.	N/A	N/A
2010/0119951	12/2009	Abouimrane et al.	N/A	N/A
2010/0140560	12/2009	Wang et al.	N/A	N/A

2010/0143769	12/2009	Lee et al.	N/A	N/A
2010/0230632	12/2009	Adamson et al.	N/A	N/A
2010/0233545	12/2009	Sano et al.	N/A	N/A
2010/0266899	12/2009	Barker et al.	N/A	N/A
2010/0283012	12/2009	Hibst et al.	N/A	N/A
2010/0285372	12/2009	Lee et al.	N/A	N/A
2010/0301281	12/2009	Hibst et al.	N/A	N/A
2010/0303701	12/2009	Faulkner	N/A	N/A
2011/0008678	12/2010	Li et al.	N/A	N/A
2011/0012067	12/2010	Kay	N/A	N/A
2011/0052473	12/2010	Sano et al.	N/A	N/A
2011/0052986	12/2010	Barker et al.	N/A	N/A
2011/0052995	12/2010	Sano et al.	N/A	N/A
2011/0084238	12/2010	Hibst et al.	N/A	N/A
2011/0086273	12/2010	Ravet et al.	N/A	N/A
2011/0097479	12/2010	Ravet et al.	N/A	N/A
2011/0104534	12/2010	Wei et al.	N/A	N/A
2011/0117417	12/2010	Pitts	N/A	N/A
2011/0123866	12/2010	Pan et al.	N/A	N/A
2011/0159381	12/2010	Doe et al.	N/A	N/A
2011/0186741	12/2010	Ohta et al.	N/A	N/A
2011/0188630	12/2010	Ohta et al.	N/A	N/A
2011/0188633	12/2010	Ohta et al.	N/A	N/A
2011/0195306	12/2010	Chiang et al.	N/A	N/A
2011/0200848	12/2010	Chiang et al.	N/A	N/A
2011/0229765	12/2010	Barker et al.	N/A	N/A
2011/0274948	12/2010	Duduta et al.	N/A	N/A
2011/0287316	12/2010	Lu et al.	N/A	N/A
2011/0317817	12/2010	Nishino et al.	N/A	N/A
2011/0318652	12/2010	Furuya	N/A	N/A
2012/0002784	12/2011	Nishino et al.	N/A	N/A
2012/0064401	12/2011	Liu et al.	N/A	N/A
2012/0100402	12/2011	Nesper et al.	N/A	N/A
2012/0138867	12/2011	Liang et al.	N/A	N/A
2012/0140378	12/2011	Tan et al.	N/A	N/A
2012/0164499	12/2011	Chiang et al.	N/A	N/A
2012/0164517	12/2011	Lee et al.	N/A	N/A
2012/0205676	12/2011	Seo et al.	N/A	N/A
2012/0205685	12/2011	Seo et al.	N/A	N/A
2012/0205686	12/2011	Seo et al.	N/A	N/A
2012/0213919	12/2011	Ravet et al.	N/A	N/A
2012/0214071	12/2011	Chiang et al.	N/A	N/A
2012/0214072	12/2011	Ravet et al.	N/A	N/A
2012/0219856	12/2011	Doe et al.	N/A	N/A
2012/0219859	12/2011	Doe et al.	N/A	N/A
2012/0219862	12/2011	Fujino et al.	N/A	N/A
2012/0237828	12/2011	Tan et al.	N/A	N/A
2012/0270107	12/2011	Toya et al.	N/A	N/A
2012/0270109	12/2011	Chiang et al.	N/A	N/A
2012/0276454	12/2011	Mori et al.	N/A	N/A

2012/0301787	12/2011	Yamamura et al.	N/A	N/A
2012/0302697	12/2011	Inada et al.	N/A	N/A
2012/0315538	12/2011	Chiang et al.	N/A	N/A
2012/0321953	12/2011	Chen	977/773	H01M 4/483
2013/0026409	12/2012	Baker et al.	N/A	N/A
2013/0029207	12/2012	Jain	N/A	N/A
2013/0034780	12/2012	Muldoon et al.	N/A	N/A
2013/0043437	12/2012	Wang et al.	N/A	N/A
2013/0048967	12/2012	Nishido et al.	N/A	N/A
2013/0052492	12/2012	Schaefer	N/A	N/A
2013/0059204	12/2012	Holzapfel et al.	N/A	N/A
2013/0059211	12/2012	Schaefer et al.	N/A	N/A
2013/0074614	12/2012	Holmes et al.	N/A	N/A
2013/0078149	12/2012	Holmes et al.	N/A	N/A
2013/0078520	12/2012	Toya et al.	N/A	N/A
2013/0078523	12/2012	Rojeski	N/A	N/A
2013/0078524	12/2012	Rojeski	N/A	N/A
2013/0078624	12/2012	Holmes et al.	N/A	N/A
2013/0078625	12/2012	Holmes et al.	N/A	N/A
2013/0078733	12/2012	Holmes et al.	N/A	N/A
2013/0079236	12/2012	Holmes	N/A	N/A
2013/0079599	12/2012	Holmes et al.	N/A	N/A
2013/0084474	12/2012	Mills	N/A	N/A
2013/0089486	12/2012	Ceder et al.	N/A	N/A
2013/0108925	12/2012	Holzapfel	N/A	N/A
2013/0115521	12/2012	Doe et al.	N/A	N/A
2013/0119358	12/2012	Sasaki et al.	N/A	N/A
2013/0137200	12/2012	Shimomura	N/A	N/A
2013/0153821	12/2012	Evenson et al.	N/A	N/A
2013/0157126	12/2012	Sulaiman Lo et al.	N/A	N/A
2013/0157135	12/2012	Zhou et al.	N/A	N/A
2013/0189590	12/2012	Jean-Marc et al.	N/A	N/A
2013/0214201	12/2012	Hibst et al.	N/A	N/A
2013/0214212	12/2012	Hibst et al.	N/A	N/A
2013/0214462	12/2012	Kikuchi et al.	N/A	N/A
2013/0216903	12/2012	Pitteloud	N/A	N/A
2013/0252112	12/2012	Doe et al.	N/A	N/A
2013/0252114	12/2012	Doe et al.	N/A	N/A
2013/0260225	12/2012	Doe et al.	N/A	N/A
2013/0260228	12/2012	Sano et al.	N/A	N/A
2013/0260238	12/2012	Doe et al.	N/A	N/A
2013/0273402	12/2012	Tsutsumi et al.	N/A	N/A
2013/0288129	12/2012	Toya et al.	N/A	N/A
2013/0300284	12/2012	Nishido	N/A	N/A
2013/0302673	12/2012	Ravet et al.	N/A	N/A
2013/0302697	12/2012	Wang et al.	N/A	N/A
2013/0309561	12/2012	Chen et al.	N/A	N/A
2013/0316250	12/2012	Khalifah et al.	N/A	N/A
2013/0323608	12/2012	Chang et al.	N/A	N/A
2013/0327249	12/2012	Winter et al.	N/A	N/A

2013/0330603	12/2012	Kroll et al.	N/A	N/A
2013/0330611	12/2012	Chen et al.	N/A	N/A
2013/0337331	12/2012	Mori et al.	N/A	N/A
2013/0344367	12/2012	Chiang et al.	N/A	N/A
2014/0011090	12/2013	Toya et al.	N/A	N/A
2014/0027743	12/2013	Nishido	N/A	N/A
2014/0056797	12/2013	Kabe et al.	N/A	N/A
2014/0072871	12/2013	Chen et al.	N/A	N/A
2014/0072879	12/2013	Chen et al.	N/A	N/A
2014/0072883	12/2013	Kondo et al.	N/A	N/A
2014/0073043	12/2013	Holmes	N/A	N/A
2014/0099557	12/2013	Doe et al.	N/A	N/A
2014/0106214	12/2013	Doe et al.	N/A	N/A
2014/0127562	12/2013	Nesper et al.	N/A	N/A
2014/0159011	12/2013	Suzuki et al.	N/A	N/A
2014/0170476	12/2013	Tan et al.	N/A	N/A
2014/0170735	12/2013	Holmes	N/A	N/A
2014/0186238	12/2013	Holmes et al.	N/A	N/A
2014/0191220	12/2013	Watabe et al.	N/A	N/A
2014/0212733	12/2013	Johnson et al.	N/A	N/A
2014/0220450	12/2013	Jilek et al.	N/A	N/A
2014/0234695	12/2013	Kondo et al.	N/A	N/A
2014/0234949	12/2013	Wasson et al.	N/A	N/A
2014/0242445	12/2013	Gozdz et al.	N/A	N/A
2014/0248544	12/2013	Xing et al.	N/A	N/A
2014/0275685	12/2013	Sanchez Valente et al.	N/A	N/A
2014/0294721	12/2013	Feng et al.	N/A	N/A
2014/0295275	12/2013	Kay	N/A	N/A
2014/0296089	12/2013	Holmes et al.	N/A	N/A
2014/0302403	12/2013	Doe et al.	N/A	N/A
2014/0308661	12/2013	Holmes et al.	N/A	N/A
2014/0315104	12/2013	Liu et al.	N/A	N/A
2014/0335415	12/2013	Tamaki et al.	N/A	N/A
2014/0335918	12/2013	Gardner et al.	N/A	N/A
2014/0363746	12/2013	He et al.	N/A	N/A
2014/0370186	12/2013	Liang et al.	N/A	N/A
2014/0370388	12/2013	Singh et al.	N/A	N/A
2014/0377660	12/2013	Fukui et al.	N/A	N/A
2015/0004486	12/2014	Moroishi et al.	N/A	N/A
2015/0030929	12/2014	Barker	N/A	N/A
2015/0037676	12/2014	Yamauchi et al.	N/A	N/A
2015/0044556	12/2014	Wang et al.	N/A	N/A
2015/0056677	12/2014	Abin et al.	N/A	N/A
2015/0069449	12/2014	Nishido	N/A	N/A
2015/0086471	12/2014	Sanchez Valente et al.	N/A	N/A
2015/0086874	12/2014	Kondo et al.	N/A	N/A
2015/0111105	12/2014	Kato et al.	N/A	N/A
2015/0112109	12/2014	Sanchez Valente et al.	N/A	N/A
2015/0125945	12/2014	Holmes et al.	N/A	N/A
2015/0132650	12/2014	Mizuno et al.	N/A	N/A

2015/0132660	12/2014	Ravet et al.	N/A	N/A
2015/0140476	12/2014	Nair et al.	N/A	N/A
2015/0155548	12/2014	Ryoshi et al.	N/A	N/A
2015/0214273	12/2014	Seo et al.	N/A	N/A
2015/0214506	12/2014	Seo et al.	N/A	N/A
2015/0221703	12/2014	Seo et al.	N/A	N/A
2015/0228977	12/2014	Toya et al.	N/A	N/A
2015/0236342	12/2014	Rojeski	N/A	N/A
2015/0236349	12/2014	Chiang et al.	N/A	N/A
2015/0245456	12/2014	Nishino et al.	N/A	N/A
2015/0263382	12/2014	Singh et al.	N/A	N/A
2015/0280259	12/2014	Anderson et al.	N/A	N/A
2015/0303474	12/2014	Barker et al.	N/A	N/A
2015/0311506	12/2014	Mizuno et al.	N/A	N/A
2015/0311520	12/2014	Mizuno et al.	N/A	N/A
2015/0311565	12/2014	Muldoon et al.	N/A	N/A
2015/0318335	12/2014	Kataishi et al.	N/A	N/A
2015/0338428	12/2014	Holmes et al.	N/A	N/A
2015/0357653	12/2014	Yoshida et al.	N/A	N/A
2015/0364753	12/2014	Chiang et al.	N/A	N/A
2015/0364761	12/2014	Fukui et al.	N/A	N/A
2015/0368717	12/2014	Holmes et al.	N/A	N/A
2015/0376817	12/2014	Xu et al.	N/A	N/A
2016/0003823	12/2015	Holmes	N/A	N/A
2016/0006028	12/2015	Nair et al.	N/A	N/A
2016/0006090	12/2015	Licht	N/A	N/A
2016/0007432	12/2015	Ohta et al.	N/A	N/A
2016/0011215	12/2015	Holmes	N/A	N/A
2016/0011225	12/2015	Holmes	N/A	N/A
2016/0025760	12/2015	Holmes	N/A	N/A
2016/0025763	12/2015	Holmes	N/A	N/A
2016/0028105	12/2015	Khalifah et al.	N/A	N/A
2016/0028114	12/2015	Pratt et al.	N/A	N/A
2016/0032361	12/2015	Holmes et al.	N/A	N/A
2016/0033544	12/2015	Holmes et al.	N/A	N/A
2016/0043146	12/2015	Uesaka et al.	N/A	N/A
2016/0054343	12/2015	Holmes et al.	N/A	N/A
2016/0069919	12/2015	Holmes et al.	N/A	N/A
2016/0069920	12/2015	Holmes et al.	N/A	N/A
2016/0069921	12/2015	Holmes et al.	N/A	N/A
2016/0077015	12/2015	Holmes et al.	N/A	N/A
2016/0079608	12/2015	Choi et al.	N/A	N/A
2016/0079609	12/2015	Choi et al.	N/A	N/A
2016/0084863	12/2015	Holmes et al.	N/A	N/A
2016/0087262	12/2015	Toya et al.	N/A	N/A
2016/0093885	12/2015	Kamata et al.	N/A	N/A
2016/0096334	12/2015	Sander et al.	N/A	N/A
2016/0099480	12/2015	Nie et al.	N/A	N/A
2016/0103123	12/2015	Holmes et al.	N/A	N/A
2016/0111720	12/2015	Arthur et al.	N/A	N/A

2016/0124009	12/2015	Wasson et al.	N/A	N/A
2016/0155993	12/2015	Nishido et al.	N/A	N/A
2016/0156019	12/2015	Kang et al.	N/A	N/A
2016/0164093	12/2015	Inoue et al.	N/A	N/A
2016/0164152	12/2015	Tan et al.	N/A	N/A
2016/0168086	12/2015	Gonzalez Martinez et al.	N/A	N/A
2016/0169880	12/2015	Holmes et al.	N/A	N/A
2016/0169923	12/2015	Holmes et al.	N/A	N/A
2016/0202588	12/2015	Bass et al.	N/A	N/A
2016/0204436	12/2015	Barker et al.	N/A	N/A
2016/0216286	12/2015	Holmes et al.	N/A	N/A
2016/0216287	12/2015	Holmes et al.	N/A	N/A
2016/0233437	12/2015	Suzuki et al.	N/A	N/A
2016/0246153	12/2015	Garcia et al.	N/A	N/A
2016/0268605	12/2015	Yamauchi et al.	N/A	N/A
2016/0276707	12/2015	Holub et al.	N/A	N/A
2016/0285086	12/2015	Zhu et al.	N/A	N/A
2016/0320381	12/2015	Holmes et al.	N/A	N/A
2016/0322589	12/2015	Watabe et al.	N/A	N/A
2016/0343949	12/2015	Seo et al.	N/A	N/A
2016/0343954	12/2015	Seo et al.	N/A	N/A
2016/0351904	12/2015	Goodenough et al.	N/A	N/A
2016/0358716	12/2015	Gardner et al.	N/A	N/A
2016/0365577	12/2015	Zhang	N/A	H01M 4/5825
2016/0370396	12/2015	Wasson et al.	N/A	N/A
2016/0372688	12/2015	Seo et al.	N/A	N/A
2016/0377640	12/2015	Balwani et al.	N/A	N/A
2016/0380236	12/2015	Seo et al.	N/A	N/A
2017/0005327	12/2016	Goodenough et al.	N/A	N/A
2017/0005332	12/2016	Chen et al.	N/A	N/A
2017/0012207	12/2016	Seo et al.	N/A	N/A
2017/0012232	12/2016	Kataishi et al.	N/A	N/A
2017/0025479	12/2016	Seo et al.	N/A	N/A
2017/0025615	12/2016	Seo et al.	N/A	N/A
2017/0025630	12/2016	Seo et al.	N/A	N/A
2017/0038401	12/2016	Holmes et al.	N/A	N/A
2017/0040553	12/2016	Watabe et al.	N/A	N/A
2017/0062869	12/2016	Zhamu et al.	N/A	N/A
2017/0069852	12/2016	Kanamoto et al.	N/A	N/A
2017/0084670	12/2016	Seo et al.	N/A	N/A
2017/0092889	12/2016	Seo et al.	N/A	N/A
2017/0092890	12/2016	Seo et al.	N/A	N/A
2017/0098856	12/2016	Zhamu et al.	N/A	N/A
2017/0104204	12/2016	Zhamu et al.	N/A	N/A
2017/0104208	12/2016	Yamauchi et al.	N/A	N/A
2017/0117524	12/2016	Chen et al.	N/A	N/A
2017/0125703	12/2016	Suzuki et al.	N/A	N/A
2017/0146882	12/2016	Bass et al.	N/A	N/A
2017/0179527	12/2016	Schmitz et al.	N/A	N/A

2017/0179558	12/2016	McKone et al.	N/A	N/A
2017/0186971	12/2016	Kanamoto et al.	N/A	N/A
2017/0186980	12/2016	Watabe et al.	N/A	N/A
2017/0213876	12/2016	Ohsawa et al.	N/A	N/A
2017/0213991	12/2016	Uesaka et al.	N/A	N/A
2017/0222172	12/2016	Watabe et al.	N/A	N/A
2017/0237118	12/2016	Khalifah et al.	N/A	N/A
2017/0244059	12/2016	Sasaki et al.	N/A	N/A
2017/0256795	12/2016	Semenenko et al.	N/A	N/A
2017/0271610	12/2016	Takahashi	N/A	N/A
2017/0283845	12/2016	Holmes et al.	N/A	N/A
2017/0309687	12/2016	Watabe et al.	N/A	N/A
2017/0309852	12/2016	Seo et al.	N/A	N/A
2017/0312746	12/2016	Holmes et al.	N/A	N/A
2017/0324054	12/2016	Ishisone et al.	N/A	N/A
2017/0324055	12/2016	Ishisone et al.	N/A	N/A
2017/0324081	12/2016	Toya et al.	N/A	N/A
2017/0338436	12/2016	Mitsumori et al.	N/A	N/A
2017/0338472	12/2016	Zhamu et al.	N/A	N/A
2017/0338485	12/2016	Toya et al.	N/A	N/A
2017/0350878	12/2016	Holmes et al.	N/A	N/A
2017/0352868	12/2016	Zhamu et al.	N/A	N/A
2017/0352869	12/2016	Zhamu et al.	N/A	N/A
2017/0352891	12/2016	Choi et al.	N/A	N/A
2017/0354955	12/2016	Hossain et al.	N/A	N/A
2017/0373310	12/2016	Whittingham et al.	N/A	N/A
2018/0034090	12/2017	Chiang et al.	N/A	N/A
2018/0040877	12/2017	Ravet et al.	N/A	N/A
2018/0045745	12/2017	Holmes et al.	N/A	N/A
2018/0053933	12/2017	Ryoshi et al.	N/A	N/A
2018/0062214	12/2017	Smith et al.	N/A	N/A
2018/0115014	12/2017	Zanotti et al.	N/A	N/A
2018/0131040	12/2017	Visco et al.	N/A	N/A
2018/0145326	12/2017	Kensaku et al.	N/A	N/A
2018/0154338	12/2017	Hossain et al.	N/A	N/A
2018/0159123	12/2017	Mizuno et al.	N/A	N/A
2018/0169421	12/2017	Chen et al.	N/A	N/A
2018/0175433	12/2017	Zhamu et al.	N/A	N/A
2018/0175434	12/2017	Zhamu et al.	N/A	N/A
2018/0183062	12/2017	Zhamu et al.	N/A	N/A
2018/0183066	12/2017	Zhamu et al.	N/A	N/A
2018/0183089	12/2017	Xu et al.	N/A	N/A
2018/0183107	12/2017	Zhamu et al.	N/A	N/A
2018/0190983	12/2017	Fukui et al.	N/A	N/A
2018/0205068	12/2017	Takeuchi et al.	N/A	N/A
2018/0205080	12/2017	Toya et al.	N/A	N/A
2018/0212241	12/2017	Liu et al.	N/A	N/A
2018/0219048	12/2017	Seo et al.	N/A	N/A
2018/0241080	12/2017	Jena et al.	N/A	N/A
2018/0248185	12/2017	Johnson et al.	N/A	N/A

2018/0248189	12/2017	Pan et al.	N/A	N/A
2018/0248190	12/2017	Pan et al.	N/A	N/A
2018/0254481	12/2017	Kamata et al.	N/A	N/A
2018/0277894	12/2017	Pan et al.	N/A	N/A
2018/0277913	12/2017	Pan et al.	N/A	N/A
2018/0287150	12/2017	Goodenough et al.	N/A	N/A
2018/0294474	12/2017	Zhamu et al.	N/A	N/A
2018/0294476	12/2017	Zhamu et al.	N/A	N/A
2018/0301707	12/2017	Pan et al.	N/A	N/A
2018/0307113	12/2017	Bass et al.	N/A	N/A
2018/0323474	12/2017	Liu et al.	N/A	N/A
2018/0342731	12/2017	Ravet et al.	N/A	N/A
2018/0342733	12/2017	Goodenough et al.	N/A	N/A
2018/0342737	12/2017	Zhamu et al.	N/A	N/A
2018/0347069	12/2017	Toya et al.	N/A	N/A
2018/0351200	12/2017	Zhamu et al.	N/A	N/A
2018/0351201	12/2017	Zhamu et al.	N/A	N/A
2018/0375156	12/2017	Zhamu et al.	N/A	N/A
2019/0006671	12/2018	Uekusa et al.	N/A	N/A
2019/0020025	12/2018	Matsumoto et al.	N/A	N/A
2019/0027542	12/2018	Watabe et al.	N/A	N/A
2019/0027788	12/2018	Liu et al.	N/A	N/A
2019/0044137	12/2018	Zhamu et al.	N/A	N/A
2019/0051905	12/2018	Zhamu et al.	N/A	N/A
2019/0067732	12/2018	Zhamu et al.	N/A	N/A
2019/0071319	12/2018	Rodriguez et al.	N/A	N/A
2019/0088882	12/2018	Seo et al.	N/A	N/A
2019/0097210	12/2018	Chen et al.	N/A	N/A
2019/0115591	12/2018	Zhamu et al.	N/A	N/A
2019/0148715	12/2018	Liu et al.	N/A	N/A
2019/0165365	12/2018	Zhamu et al.	N/A	N/A
2019/0165374	12/2018	Zhamu et al.	N/A	N/A
2019/0173079	12/2018	Zhamu et al.	N/A	N/A
2019/0173082	12/2018	Zhamu et al.	N/A	N/A
2019/0180951	12/2018	Yokota et al.	N/A	N/A
2019/0189936	12/2018	Watabe et al.	N/A	N/A
2019/0190022	12/2018	Kim et al.	N/A	N/A
2019/0207124	12/2018	Kanamoto et al.	N/A	N/A
2019/0207200	12/2018	Zhamu et al.	N/A	N/A
2019/0259583	12/2018	Baba et al.	N/A	N/A
2019/0273257	12/2018	Siu et al.	N/A	N/A
2019/0305290	12/2018	Singh et al.	N/A	N/A
2019/0312272	12/2018	Goodenough et al.	N/A	N/A
2019/0341652	12/2018	Laughman et al.	N/A	N/A
2019/0348625	12/2018	Mitsumori et al.	N/A	N/A
2019/0355963	12/2018	Singh et al.	N/A	N/A
2019/0363394	12/2018	Zhamu et al.	N/A	N/A
2019/0372093	12/2018	Zhamu et al.	N/A	N/A
2019/0372099	12/2018	Zhamu et al.	N/A	N/A
2019/0372100	12/2018	Zhamu et al.	N/A	N/A

2019/0372148	12/2018	He et al.	N/A	N/A
2019/0372151	12/2018	Zhamu et al.	N/A	N/A
2019/0372174	12/2018	He et al.	N/A	N/A
2019/0372449	12/2018	Mills	N/A	N/A
2019/0379021	12/2018	He et al.	N/A	N/A
2019/0379039	12/2018	Zhamu et al.	N/A	N/A
2019/0379045	12/2018	He et al.	N/A	N/A
2019/0393467	12/2018	Guo et al.	N/A	N/A
2019/0393482	12/2018	He et al.	N/A	N/A
2019/0393485	12/2018	He et al.	N/A	N/A
2019/0393486	12/2018	He et al.	N/A	N/A
2019/0393487	12/2018	He et al.	N/A	N/A
2019/0393495	12/2018	He et al.	N/A	N/A
2019/0393496	12/2018	He et al.	N/A	N/A
2019/0393508	12/2018	Zhamu et al.	N/A	N/A
2019/0393541	12/2018	Jang	N/A	N/A
2019/0393542	12/2018	Jang	N/A	N/A
2019/0393543	12/2018	Zhamu et al.	N/A	N/A
2020/0006770	12/2019	Kamata et al.	N/A	N/A
2020/0014057	12/2019	Xu et al.	N/A	N/A
2020/0020895	12/2019	Collins et al.	N/A	N/A
2020/0028147	12/2019	Zhamu et al.	N/A	N/A
2020/0028158	12/2019	Zhamu et al.	N/A	N/A
2020/0028205	12/2019	Zhamu et al.	N/A	N/A
2020/0044250	12/2019	Toya et al.	N/A	N/A
2020/0052042	12/2019	Seo et al.	N/A	N/A
2020/0052325	12/2019	Zhamu et al.	N/A	N/A
2020/0052350	12/2019	Zhamu et al.	N/A	N/A
2020/0057085	12/2019	Wasson et al.	N/A	N/A
2020/0058936	12/2019	Horikawa et al.	N/A	N/A
2020/0067077	12/2019	Pan et al.	N/A	N/A
2020/0067079	12/2019	Pan et al.	N/A	N/A
2020/0067080	12/2019	Pan et al.	N/A	N/A
2020/0067101	12/2019	Pan et al.	N/A	N/A
2020/0083543	12/2019	Choi et al.	N/A	N/A
2020/0091507	12/2019	Zhamu et al.	N/A	N/A
2020/0098997	12/2019	Watabe et al.	N/A	N/A
2020/0144595	12/2019	Lin	N/A	N/A
2020/0144600	12/2019	Zhong et al.	N/A	N/A
2020/0144605	12/2019	Su et al.	N/A	N/A
2020/0153068	12/2019	Licht	N/A	N/A
2020/0207881	12/2019	Ohkimoto et al.	N/A	N/A
2020/0230596	12/2019	Holmes et al.	N/A	N/A
2020/0235421	12/2019	Zhou et al.	N/A	N/A
2020/0243838	12/2019	Jang	N/A	N/A
2020/0243854	12/2019	Jang	N/A	N/A
2020/0253047	12/2019	Lai et al.	N/A	N/A
2020/0259105	12/2019	Watabe et al.	N/A	N/A
2020/0259200	12/2019	Nie et al.	N/A	N/A
2020/0266426	12/2019	Zhamu et al.	N/A	N/A

2020/0280012	12/2019	Mitsumori et al.	N/A	N/A
2020/0280054	12/2019	Jang	N/A	N/A
2020/0280055	12/2019	Jang	N/A	N/A
2020/0287206	12/2019	Jang	N/A	N/A
2020/0287207	12/2019	Jang	N/A	N/A
2020/0295267	12/2019	Kanamoto et al.	N/A	N/A
2020/0295355	12/2019	Kanno et al.	N/A	N/A
2020/0295399	12/2019	Zhou et al.	N/A	N/A
2020/0313174	12/2019	Wang et al.	N/A	N/A
2020/0321614	12/2019	Banerjee et al.	N/A	N/A
2020/0326356	12/2019	Wasson et al.	N/A	N/A
2020/0350373	12/2019	Ohsawa et al.	N/A	N/A
2020/0350503	12/2019	Seo et al.	N/A	N/A
2020/0350508	12/2019	Seo et al.	N/A	N/A
2020/0350589	12/2019	Jang	N/A	N/A
2020/0354220	12/2019	Pedersen et al.	N/A	N/A
2020/0358090	12/2019	Jang	N/A	N/A
2020/0358141	12/2019	Jang	N/A	N/A
2020/0365902	12/2019	Jang	N/A	N/A
2020/0366180	12/2019	Mills	N/A	N/A
2020/0381728	12/2019	Uekusa et al.	N/A	N/A
2020/0395566	12/2019	Ishisone et al.	N/A	N/A
2020/0403555	12/2019	Mills	N/A	N/A
2021/0021001	12/2020	Chang et al.	N/A	N/A
2021/0021003	12/2020	Chen et al.	N/A	N/A
2021/0028451	12/2020	Johnson et al.	N/A	N/A
2021/0028507	12/2020	Su et al.	N/A	N/A
2021/0028509	12/2020	Su et al.	N/A	N/A
2021/0043929	12/2020	Huang et al.	N/A	N/A

OTHER PUBLICATIONS

Berrah, Fadila, et al. "The vanadium monophosphates AVOPO₄: Synthesis of a second form β -KVOPO₄ and structural relationships in the series." *Solid state sciences* 3.4 (2001): 477-482. cited by applicant

Zima, Vítězslav, et al. "Ion-exchange properties of alkali-metal redox-intercalated vanadyl phosphate." *Journal of Solid State Chemistry* 163.1 (2002): 281-285. cited by applicant

Lii, Kwang-Hwa, and Wei-Chuan Liu. "RbVOPO₄ and CsVOPO₄, Two Vanadyl (IV) Orthophosphates with an Intersecting Tunnel Structure and Discrete VO₅ Pyramids." *Journal of Solid State Chemistry* 103.1 (1993): 38-44. cited by applicant

Yakubovich, O. V., O. V. Karimova, and O. K. Mel'nikov. "The mixed anionic framework in the structure of Na₂ {MnF [PO₄]}. " *Acta Crystallographica Section C: Crystal Structure Communications* 53.4 (1997): 395-397. cited by applicant

Schindler, M., F. C. Hawthorne, and W. H. Baur. "Crystal chemical aspects of vanadium: polyhedral geometries, characteristic bond valences, and polymerization of (VO n) polyhedra." *Chemistry of Materials* 12.5 (2000): 1248-1259. cited by applicant

Panin, Rodion V., et al. "Crystal Structure, Polymorphism, and Properties of the New Vanadyl Phosphate Na₄VO (PO₄)₂." *Chemistry of materials* 16.6 (2004): 1048-1055. cited by applicant

Belkhiri, Sabrina, Djillali Mezaoui, and Thierry Roisnel. "The Structure Determination of a New Mixed Mono-Arsenate K₂V₂O₂ (AsO₄)₂." *3ème Conférence Internationale sur le Soudage, le CND et l'Industrie des Matériaux et Alliages (IC-WNDT-MI'12)*. Centre de Recherche Scientifique

et Technique en Soudage et Contrôle (CSC), 2012 cited by applicant

Glasser, Leslie, and C. Richard A. Catlow. "Modelling phase changes in the potassium titanyl phosphate system." *Journal of Materials Chemistry* 7.12 (1997): 2537-2542. cited by applicant

Fedotov, Stanislav S., et al. "AVPO₄F (A=Li, K): A 4 V Cathode Material for High-Power Rechargeable Batteries." *Chemistry of Materials* 28.2 (2016): 411-415. cited by applicant

Belkhiri, Sabrina, Djillali Mezaoui, and Thierry Roisnel. "K₂V₂O₂ (AsO₄)₂." *Acta Crystallographica Section E: Structure Reports Online* 68.7 (2012): i54-i54. cited by applicant

Yakubovich, O. V., V. V. Kireev, and O. K. Mel'nikov. "Refinement of crystal structure of a Ge-analogue of natisite Na₂ {TiGeO₄} and prediction of new phases with anionic {MTO₅} radicals." *Crystallography Reports* 45.4 (2000): 578-584. cited by applicant

Boudin, S., et al. "Review on vanadium phosphates with mono and divalent metallic cations: syntheses, structural relationships and classification, properties." *International Journal of Inorganic Materials* 2.6 (2000): 561-579. cited by applicant

Kerr, T. A., J. Gaubicher, and L. F. Nazar. "Highly Reversible Li Insertion at 4 V in ϵ -VOPO₄/ α -LiVOPO₄ Cathodes." *Electrochemical and Solid-State Letters* 3, No. 10 (2000): 460-462. cited by applicant

Quackenbush, Nicholas F., Linda Wangoh, Bohua Wen, Ruibo Zhang, Youngmin Chung, Natasha Chernova, Zehua Chen et al. "Interfacial Effects of Electrochemical Lithiation of Epsilon-VOPO₄ and Evolution of the Electronic Structure." In *Meeting Abstracts*, No. 6, pp. 491-491. The Electrochemical Society, 2015. cited by applicant

Whittingham, M. Stanley. "Lithium batteries and cathode materials." *Chemical reviews* 104, No. 10 (2004): 4271-4302. cited by applicant

Winter, Martin, and Ralph J. Brodd. "What are batteries, fuel cells, and supercapacitors ?." *Chemical reviews* 104, No. 10 (2004): 4245-4270. cited by applicant

Melot, Brent C., and J-M. Tarascon. "Design and preparation of materials for advanced electrochemical storage." *Accounts of chemical research* 46, No. 5 (2013): 1226-1238. cited by applicant

Whittingham, M. Stanley. "Ultimate limits to intercalation reactions for lithium batteries." *Chemical reviews* 114, No. 23 (2014): 11414-11443. cited by applicant

Zaghib, K., A. Mauger, F. Gendron, and C. M. Julien. "Surface effects on the physical and electrochemical properties of thin LiFePO₄ particles." *Chemistry of Materials* 20, No. 2 (2008): 462-469. cited by applicant

Liu, Hao, Fiona C. Strobridge, Olaf J. Borkiewicz, Kamila M. Wiaderek, Karena W. Chapman, Peter J. Chupas, and Clare P. Grey. "Capturing metastable structures during high-rate cycling of LifePO₄ nanoparticle electrodes." *Science* 344, No. 6191 (2014). cited by applicant

Hautier, Geoffroy, Anubhav Jain, Shyue Ping Ong, Byoungwoo Kang, Charles Moore, Robert Doe, and Gerbrand Ceder. "Phosphates as lithium-ion battery cathodes: an evaluation based on high-throughput ab initio calculations." *Chemistry of Materials* 23, No. 15 (2011): 3495-3508. cited by applicant

Siu, Carrie, Ieuan D. Seymour, Sylvia Britto, Hanlei Zhang, Jatinkumar Rana, Jun Feng, Fredrick O. Omenya et al. "Enabling multi-electron reaction of ϵ -VOPO₄ to reach theoretical capacity for lithium-ion batteries." *Chemical communications* 54, No. 56 (2018): 7802-7805. cited by applicant

Lim, S. C., J. T. Vaughey, W. T. A. Harrison, L. L. Dussack, A. J. Jacobson, and J. W. Johnson. "Redox transformations of simple vanadium phosphates: the synthesis of ϵ -VOPO₄." *Solid State Ionics* 84, No. 3-4 (1996): 219-226. cited by applicant

Kerr, T. A., J. Gaubicher, and L. F. Nazar. "Highly Reversible Li Insertion at 4 V in ϵ -VOPO₄/ α -LiVOPO₄ Cathodes." *Electrochemical and Solid State Letters* 3, No. 10 (2000): 460. cited by applicant

Song, Yanning, Peter Y. Zavalij, and M. Stanley Whittingham. " ϵ -VOPO₄: electrochemical synthesis and enhanced cathode behavior." *Journal of the Electrochemical Society* 152, No. 4

(2005): A721. cited by applicant
Lin, Yuh-Chieh, Bohua Wen, Kamila M. Wiaderek, Shawn Sallis, Hao Liu, Saul H. Lapidus, Olaf J. Borkiewicz et al. "Thermodynamics, kinetics and structural evolution of ϵ -LiVOPO₄ over multiple lithium intercalation." Chemistry of Materials 28, No. 6 (2016): 1794-1805. cited by applicant
Chen, Zehua, Qiyuan Chen, Liquan Chen, Ruibo Zhang, Hui Zhou, Natasha A. Chernova, and M. Stanley Whittingham. "Electrochemical Behavior of Nanostructured ϵ -VOPO₄ over Two Redox Plateaus." Journal of The Electrochemical Society 160, No. 10 (2013): A1777. cited by applicant
Azmi, Bustam M., Hasanaly S. Munirah, Tatsumi Ishihara, and Yusaku Takita. "Optimized LiVOPO₄ for cathodes in Li-ion rechargeable batteries." Ionics 11, No. 5-6 (2005): 402-405. cited by applicant
Bianchini, Mateos, J. M. Ateba-Mba, Philippe Dagault, Elena Bogdan, D. Carlier, Emmanuelle Suard, Christian Masquelier, and Laurence Croguennec. "Multiple phases in the ϵ -VPO₄O—Li₂VPO₄O system: a combined solid state electrochemistry and diffraction structural study." Journal of Materials Chemistry A 2, No. 26 (2014): 10182-10192. cited by applicant
Harrison, Katharine L., Craig A. Bridges, Carlo U. Segre, C. Daniel Varnado Jr, Danielle Applestone, Christopher W. Bielawski, Mariappan Parans Paranthaman, and Arumugam Manthiram. "Chemical and electrochemical lithiation of LiVOPO₄ cathodes for lithium-ion batteries." Chemistry of Materials 26, No. 12 (2014): 3849-3861. cited by applicant
Chernova, Natasha A., Megan Roppolo, Anne C. Dillon, and M. Stanley Whittingham. "Layered vanadium and molybdenum oxides: batteries and electrochromics." Journal of Materials Chemistry 19, No. 17 (2009): 2526-2552. cited by applicant

Primary Examiner: Raymond; Brittany L

Attorney, Agent or Firm: Hoffberg & Associates

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) The present invention is a Continuation of U.S. patent application Ser. No. 18/354,493, filed Jul. 18, 2023, now U.S. Pat. No. 12,029,957, issued Jun. 4, 2024, which is a Continuation of U.S. patent application Ser. No. 17/670,470, filed Feb. 13, 2022, now U.S. Pat. No. 11,715,929, issued Aug. 1, 2023, which is a Division of U.S. patent application Ser. No. 16/291,617, filed Mar. 4, 2019, now U.S. Pat. No. 11,251,430, issued Feb. 15, 2022, which is a Non-Provisional of and claims benefit of priority from, U.S. Patent Application No. 62/638,893, filed Mar. 5, 2018, the entirety of which is expressly incorporated herein by reference.

FIELD OF THE INVENTION

(1) The present invention relates to the field of electrode materials for batteries, and more particularly to an ϵ -vanadyl phosphate cathode having near theoretical dual lithium ion capacity.

BACKGROUND OF THE INVENTION

(2) Energy conversion and storage have become more and more important in transportation, commercial, industrial, residential, and consumer applications. In particular, large-scale implementation of renewable energy, increasing ubiquity of portable electronics, and the next generation of electric vehicles require inexpensive and efficient energy storage systems.

(3) A lithium secondary battery includes a cathode, an anode, and a separator. The separator may be a solid electrolyte, or an additional element with a liquid electrolyte. During discharging of the lithium secondary battery, oxidation reaction occurs in the anode due to deintercalation of lithium

ions, while reduction reaction occurs in the cathode due to intercalation of lithium ions. The vice versa processes take place during the battery charging. The electrolyte has selective conductivity only ions, and thus transfers lithium ions between the cathode and the anode. Lithium ions intercalated into an electrode of a battery lead to charge neutrality with electrons entered into the electrode, and thus serve as media storing electric energy in the electrode. Accordingly, the quantity of electric energy storable by the battery is dependent upon the quantity of lithium ions intercalated into the electrode. Although basic performance of the lithium secondary battery, such as operating voltage and energy density, is dependent upon the materials of the cathode and anode, the electrolyte also needs to have high-ion conductivity, electrochemical stability and thermal stability to ensure high performance of the lithium secondary battery.

(4) A typical lithium-ion battery electrolyte consists of a lithium salt and a solvent. Because of the high operating voltage, the solvent is typically anhydrous, with organic solvents now common, e.g., glyme. Phosphazenes and phosphoranimines have been proposed as an alternate non-flammable electrolyte. Other nitrogen, sulfur, phosphorus, silicon, compounds are also known as electrolyte additives or electrolytes. The electrolyte needs to be electrochemically stable in a voltage range where reduction and oxidation proceed in the anode and cathode, respectively.

(5) As the use of lithium secondary batteries is expanding to electric vehicles and power storage fields, electrode active materials for use at high voltages emerged and became available. Use of a relatively low-potential anode active material and a relatively high-potential cathode active material has led to a narrower potential window of the electrolyte, so that the electrolyte is more likely to decompose on a surface of the cathode/anode. Lithium secondary batteries for electric vehicles and power storage are likely to be exposed to external high-temperature environment conditions, and the temperatures of these lithium secondary batteries may rise during instantaneous charging and discharging. Accordingly, lifetime and stored energy quantity of the lithium secondary battery may be reduced in such high-temperature environment conditions.

(6) Typical lithium secondary battery technologies are discussed in U.S. Pat. No. 9,819,054, expressly incorporated herein by reference in its entirety.

(7) The non-aqueous solvent, which is in the electrolyte of a lithium secondary battery according to the above-described embodiments, may serve as a migration medium of ions involved in electrochemical reactions of the battery. Any suitable non-aqueous solvent that is commonly used in the art may be used. For example, the non-aqueous solvent may be an organic carbonate compound, an ester compound, an ether compound, a ketone compound, an alcohol compound, an aprotic bipolar solvent, or a combination thereof. The carbonate compound may be an open chain carbonate compound, a cyclic carbonate compound, a fluoro carbonate derivative thereof, or a combination thereof.

(8) The electrolyte useful for the battery is one which does not chemically react with the anode or with the cathode during storage, and permits the migration of ions to intercalate the cathode-active material and vice-versa (during the discharge and charging cycles, respectively). The electrolyte may be present in a pure state (in the form of a solid, fused solid or liquid) or it may be conveniently dissolved in a suitable solvent. As a general rule, the electrolyte material should consist of a compound of the same species as that which is selected for the anode-active material. Thus, useful electrolytes may be conveniently represented by the general formula LY wherein L is a cationic moiety selected from the same materials useful as the anode-active material and Y is an anionic moiety or moieties such as halides, sulfates, nitrates, beta-aluminas, phosphofluorides, perchlorates and rubidium halide. The electrolyte may be present in a pure state in the form of a solid, fused solid (i.e. molten salt) or liquid or it may be conveniently dissolved in a suitable solvent which does not generally hydrolyze or degrade under conditions within the battery. Such electrolytes include ketones, esters, ethers, organic carbonates (such as propylene carbonate), organic lactones, organic nitriles, nitrohydrocarbons, organic sulfoxides, etc. and mixtures thereof. Where the solvent is utilized, the electrolyte salt may be present in a concentration determined by

the desired solution conductivity, solubility and chemical reactivity. The electrolyte may include additives to reduce flammability, such as phosphazenes, e.g., cyclic phosphazenes.

(9) Non-limiting examples of the chain carbonate compound are diethyl carbonate (“DEC”), dimethyl carbonate, (“DMC”), dipropyl carbonate (“DPC”), methylpropyl carbonate (“MPC”), ethylpropylcarbonate (“EPC”), methylethyl carbonate (“MEC”), and a combination thereof. Non-limiting examples of the cyclic carbonate compound are ethylene carbonate (“EC”), propylenecarbonate (“PC”), butylene carbonate (“BC”), fluoroethylene carbonate (“FEC”), vinylene carbonate (“VEC”), and a combination thereof. Non-limiting examples of the fluorocarbonate compound are fluoroethylene carbonate (“FEC”), 4,5-difluoroethylene carbonate, 4,4-difluoroethylene carbonate, 4,4,5-trifluoroethylene carbonate, 4,4,5,5-tetrafluoroethylene carbonate, 4-fluoro-5-methylethylene carbonate, 4-fluoro-4-methylethylene carbonate, 4,5-difluoro-4-methylethylene carbonate, 4,4,5-trifluoro-5-methylethylene carbonate, trifluoromethylethylene carbonate, and a combination thereof. The carbonate compound may include a combination of cyclic carbonate and chain carbonate, in consideration of dielectric constant and viscosity of the electrolyte. For example, an amount of a cyclic carbonate compound may be at least 10% by volume based on a total volume of the non-aqueous organic solvent. The carbonate compound may be a mixture of such chain carbonate and/or cyclic carbonate compounds as described above with a fluorocarbonate compound. The fluorocarbonate compound may increase solubility of a lithium salt to improve ionic conductivity of the electrolyte, and may facilitate formation of the thin film on the anode. In some embodiments, the fluorocarbonate compound may be fluoroethylene carbonate (“FEC”). An amount of the fluorocarbonate compound may be from about 1 to about 30 percent by volume (“volume %”) based on a total volume of the non-aqueous organic solvent. When the amount of the fluorocarbonate compound is within this range, the electrolyte may have an appropriate viscosity to provide desired effects thereof.

(10) Non-limiting examples of the ester compound are methyl acetate, ethyl acetate, n-propyl acetate, dimethyl acetate, methyl propionate (“MP”), ethyl propionate, γ -butyrolactone, decanolide, valerolactone, mevalonolactone, caprolactone, and methyl formate. Non-limiting examples of the ether compound are dibutyl ether, tetraglyme, diglyme, 1,2-dimethoxyethane, 1,2-diethoxyethane, ethoxymethoxyethane, 2-methyltetrahydrofuran, and tetrahydrofuran. An example of the ketone compound is cyclohexanone. Non-limiting examples of the alcohol compound are ethyl alcohol and isopropyl alcohol.

(11) Examples of the aprotic solvent are nitriles (such as $R-CN$, wherein R is a C.sub.2-C.sub.20 linear, branched, or cyclic hydrocarbon-based moiety that may include a double-bond, an aromatic ring or an ether bond), amides (such as formamide and dimethylformamide), dioxolanes (such as 1,2-dioxolane and 1,3-dioxolane), methylsulfoxide, sulfolanes (such as sulfolane and methylsulfolane), 1,3-dimethyl-2-imidazolidinone, N-methyl-2-pyrrolidinone, nitromethane, trimethyl phosphate, triethyl phosphate, trioctyl phosphate, and triester phosphate.

(12) The non-aqueous organic solvent may be used alone or in a combination of at least two solvents. In the latter case, a mixing ratio of the at least two non-aqueous organic solvents may be appropriately adjusted depending on a desired performance of the battery.

(13) The non-aqueous organic solvent may further include an aromatic hydrocarbon organic solvent in the carbonate solvent. The carbonate solvent and the aromatic hydrocarbon organic solvent may be mixed, for example, in a volume ratio of about 1:1 to about 30:1. Examples of the aromatic hydrocarbon organic solvent are benzene, fluorobenzene, 1,2-difluorobenzene, 1,3-difluorobenzene, 1,4-difluorobenzene, 1,2,3-trifluorobenzene, 1,2,4-trifluorobenzene, chlorobenzene, 1,2-dichlorobenzene, 1,3-dichlorobenzene, 1,4-dichlorobenzene, 1,2,3-trichlorobenzene, 1,2,4-trichlorobenzene, iodobenzene, 1,2-diiodobenzene, 1,3-diiodobenzene, 1,4-diiodobenzene, 1,2,3-triiodobenzene, 1,2,4-triiodobenzene, 2-fluorotoluene, 3-fluorotoluene, 4-fluorotoluene, 2,3-difluorotoluene, 2,4-difluorotoluene, 2,5-difluorotoluene, 2,6-difluorotoluene, 3,4-difluorotoluene, 3,5-difluorotoluene, 2,3,4-trifluorotoluene, 2,3,5-trifluorotoluene, 2,3,6-

trifluorotoluene, 3,4,5-trifluorotoluene, 2,4,5-trifluorotoluene, 2,4,6-trifluorotoluene, 2-chlorotoluene, 3-chlorotoluene, 4-chlorotoluene, 2,3-dichlorotoluene, 2,4-dichlorotoluene, 2,5-dichlorotoluene, 2,6-dichlorotoluene, 2,3,4-trichlorotoluene, 2,3,5-trichlorotoluene, 2,3,6-trichlorotoluene, 3,4,5-trichlorotoluene, 2,4,5-trichlorotoluene, 2,4,6-trichlorotoluene, 2-iodotoluene, 3-iodotoluene, 4-iodotoluene, 2,3-diiodotoluene, 2,4-diiodotoluene, 2,5-diiodotoluene, 2,6-diiodotoluene, 3,4-diiodotoluene, 3,5-diiodotoluene, 2,3,4-triiodotoluene, 2,3,5-triiodotoluene, 2,3,6-triiodotoluene, 3,4,5-triiodotoluene, 2,4,5-triiodotoluene, 2,4,6-triiodotoluene, o-xylene, m-xylene, p-xylene, and combinations thereof.

(14) Suitable electrolyte salts include, for example, LiPF_6 , LiBF_4 , LiClO_4 , LiTFSI . Suitable solvents may be any solvent which is stable within the electrochemical window of the cell and is inert to other components of the cell. Examples of suitable solvents include carbonate solvents such as ethylene carbonate, diethyl carbonate, and propylene carbonate, organic ethers such as dioxolane, dimethyl ether and tetrahydrofuran and organic nitriles such as acetonitrile.

(15) Additionally, the electrolyte may be a nonaqueous polymer electrolyte such as a gel polymer electrolyte, a solid ceramic electrolyte. In one embodiment, the electrolyte may include additives such as fluoroethylene carbonate (FEC) in order to, for example, improve cycling.

(16) The lithium salt may be any suitable lithium salt that is commonly used for lithium batteries. Examples of the lithium salt for the non-aqueous electrolyte are LiPF_6 , LiBF_4 , LiSbF_6 , LiAsF_6 , LiSbF_6 , LiCF_3SO_3 , $\text{Li}(\text{CF}_3\text{SO}_2)_2$, $\text{Li}(\text{CF}_3\text{SO}_2)_2\text{N}$, $\text{LiC}_4\text{F}_9\text{SO}_3$, LiClO_4 , LiAlO_4 , LiAlCl_4 , LiAlF_4 , LiBPh_4 , LiBiOCl , $\text{CH}_3\text{SO}_3\text{Li}$, $\text{C}_4\text{F}_9\text{SO}_3\text{Li}$, $(\text{CF}_3\text{SO}_2)_2\text{NLi}$, $\text{LiN}(\text{C}_x\text{F}_{2x+1}\text{SO}_2)(\text{C}_y\text{F}_{2y+1}\text{SO}_2)$ (wherein x and y are natural numbers), $\text{CF}_3\text{CO}_2\text{Li}$, LiCl , LiBr , LiI , LiBOB (lithium bisoxalato borate), lower aliphatic carboxylic acid lithium, lithium terphenylborate, lithium imide, and a combination thereof. These lithium salts may be used as a supporting electrolytic salt.

(17) A concentration of the lithium salt may be within a range known to one of ordinary skill in the art. The concentration of the lithium salt is not specifically limited, and in some embodiments, may be in a range of about 0.1 molar ("M") to about 2.0 M in the electrolyte to improve practical performance of a lithium battery. When the concentration of the lithium salt is within this range, the electrolyte may have appropriate conductivity and appropriate viscosity for improved performance, and may improve mobility of lithium ions.

(18) In some embodiments, the electrolyte for a lithium battery may further include an additive to facilitate the formation of a stable solid electrolyte interphase ("SEI") or a thin film on a surface of an electrode to provide improved cycle characteristics.

(19) Non-limiting examples of the additive are tris(trimethylsilyl)phosphate ("TMSPa"), lithium difluoro oxalate borate ("LiFOB"), vinylene carbonate ("VC"), propane sulfone ("PS"), succinonitrile ("SN"), LiBF_4 , a silane compound having a functional group able to form a siloxane bond (for example, acryl, amino, epoxy, methoxy, ethoxy, or vinyl), and a silazane compound such as hexamethyldisilazane. These additives may be used alone or in a combination of at least two thereof.

(20) An amount of the additive may be from about 0.01 wt % to about 10 wt % based on a total weight of the non-aqueous organic solvent. For example, the amount of the additive may be from 0.05 wt % to about 10 wt %, in some embodiments, from about 0.1 wt % to about 5 wt %, and in some other embodiments, from about 0.5 wt % to about 4 wt %, based on the total weight of the non-aqueous organic solvent. However, the amount of the additive is not particularly limited unless the additive significantly hinders improvement in capacity retention rate of a lithium battery including the electrolyte. The lithium battery may be manufactured using a method known in the art.

(21) According to the above-embodiments of the present disclosure, the lithium secondary battery may have a thin film formed on the surface of the cathode due to oxidation of at least a part of the

additive in the electrolyte during initial charging of the lithium secondary battery. Thus, the lithium secondary battery may have improved capacity retention characteristics, lifetime characteristics and high-rate characteristics even when charged at a high operating voltage of about 4.0 V to about 5.5 V, for example, a voltage about 4.3 V to about 5.5 V. An additive in the electrolyte may enhance formation of a thin film on a surface of the cathode, the thin film having a thickness of, for example, about 0.05 nanometers (“nm”) to about 100 nm. For example, the thin film may have a thickness of about 0.1 nm to about 80 nm, and in some embodiments, about 0.5 nm to about 50 nm. The thin film on the cathode surface may effectively prevent oxidation of the electrolyte on the cathode surface so that the conduction of lithium ions is not impeded.

(22) FIG. 10 is an exploded perspective view of a lithium secondary battery 100 according to an embodiment. Although the lithium secondary battery 100 illustrated in FIG. 10 is cylindrical, embodiments of the present disclosure are not limited thereto, and lithium secondary batteries according to embodiments may be of a rectangular type or a pouch type. Lithium secondary batteries may be classified as lithium-ion batteries, lithium-ion polymer batteries, or lithium polymer batteries, according to the type of separator and/or electrolyte included therein. In addition, lithium batteries may be classified as cylindrical type, rectangular type, coin type, or pouch type, according to the shape thereof. Lithium batteries may also be classified as either bulk type or thin film type, according to the size thereof. Lithium secondary batteries according to embodiments may have any appropriate shape. The structure of a lithium secondary battery and a method of manufacturing the same are known in the art, so a detailed description thereof will not be recited here. Referring to FIG. 10, the lithium secondary battery 100, which is cylindrical, includes an anode 112, a cathode 114, a separator 113 disposed between the anode 112 and the cathode 114, and an electrolyte (not shown) impregnated into the anode 112, the cathode 114 and the separator 113, a battery case 120, and a sealing member 140 sealing the battery case 120. The lithium secondary battery 100 is manufactured by sequentially stacking the anode 112, the cathode 114, and the separator 113 upon one another to form a stack, rolling the stack in a spiral form, and accommodating the rolled up stack in the battery case 120. The cathode 114 includes a cathode current collector, and a cathode active material layer disposed on the cathode current collector. The cathode current collector may have a thickness of about 3 μm to about 500 μm . The cathode current collector is not particularly limited, and may be formed of any material so long as it has a suitable conductivity without causing chemical changes in the fabricated battery. Examples of the cathode current collector include copper, stainless steel, aluminum, nickel, titanium, sintered carbon, copper or stainless steel that is surface-treated with carbon, nickel, titanium or silver, and aluminum-cadmium alloys. In addition, the cathode current collector may be processed to have fine irregularities on surfaces thereof so as to enhance adhesive strength of the cathode current collector to the cathode active material, and may be used in any of various forms including films, sheets, foils, nets, porous structures, foams, and non-woven fabrics.

(23) Known cathode active material layers include a cathode active material, a binder, and a conducting agent.

(24) Known cathode active materials include lithium-containing metal oxides, e.g., at least one of a composite oxide of lithium with a metal selected from Co, Mn, Ni, and a combination thereof.

(25) During initial cycling, a solid electrolyte interphase layer (SEI layer) forms in an electrolyte battery, representing insoluble breakdown products of the electrolyte in combination with other battery components, such as electrode material. The SEI layer serves to protect the electrolyte from further free radical reactions during overvoltage periods, e.g., during charging.

(26) The compounds listed above as cathode active materials may have a surface coating layer (hereinafter, “coating layer”). Alternatively, a mixture of a compound without a coating layer and a compound having a coating layer, the compounds being selected from the compounds listed above, may be used. The coating layer may include at least one compound of a coating element selected from oxide, hydroxide, oxyhydroxide, oxycarbonate, and hydroxycarbonate of the coating element.

The compounds for the coating layer may be amorphous or crystalline. The coating element for the coating layer may be magnesium (Mg), aluminum (Al), cobalt (Co), potassium (K), sodium (Na), calcium (Ca), silicon (Si), titanium (Ti), vanadium (V), tin (Sn), germanium (Ge), gallium (Ga), boron (B), arsenic (As), zirconium (Zr), or mixtures thereof. The coating layer may be formed using any method that does not adversely affect the physical properties of the cathode active material when a compound of the coating element is used. For example, the coating layer may be formed using a spray coating method, a dipping method, or any other method known to one of ordinary skill in the art.

(27) The binder strongly binds positive cathode active material particles together and to a current collector. Examples of the binder are, but not limited to, polyvinyl alcohol, carboxymethyl cellulose, hydroxypropyl cellulose, diacetyl cellulose, polyvinyl chloride, carboxylated polyvinyl chloride, polyvinyl fluoride, a polymer including ethylene oxide, polyvinylpyrrolidone, polyurethane, polytetrafluoroethylene, polyvinylidene fluoride, polyethylene, polypropylene, styrene-butadiene rubber (“SBR”), acrylated SBR, epoxy resin, and nylon.

(28) Known electrodes include a conducting agent used to provide conductivity to electrodes. Any electron conducting material that does not induce chemical change in batteries may be used. Examples of the conducting agent include natural graphite, artificial graphite, carbon black, acetylene black, ketjen black, carbon fibers, and metallic materials, including copper, nickel, aluminum, and silver, in powder or fiber form. The conducting agent may include a single conductive material, such as a polyphenylene derivative, or a combination of at least two conductive materials.

(29) The amounts of the cathode active material, the binder, and the conducting agent may be equivalent to those commonly used in lithium batteries. For example, a weight ratio of the cathode active material to a mixture of the conducting agent and the binder may be from about 98:2 to about 92:8, and in some embodiments from about 95:5 to about 90:10. A mixing ratio of the conducting agent to the binder may be, but not limited, from about 1:1.5 to about 1:3. The known cathode active materials may have, for example, an operating voltage range of about 4.0 V to ≥ 5.5 V.

(30) An exemplary lithium secondary battery **100**, shown in FIG. **10**, is cylindrical, and includes an anode **112**, a cathode **114**, a separator **113** disposed between the anode **112** and the cathode **114**, and an electrolyte (not shown) impregnated into the anode **112**, the cathode **114** and the separator **113**, a battery case **120**, and a sealing member **140** sealing the battery case **120**. The lithium secondary battery **100** is manufactured by sequentially stacking the anode **112**, the cathode **114**, and the separator **113** upon one another to form a stack, rolling the stack in a spiral form, and accommodating the rolled up stack in the battery case **120**.

(31) The cathode **114** includes a cathode current collector, and a cathode active material layer disposed on the cathode current collector. The cathode current collector may have a thickness of about 3 μm to about 500 μm . The cathode current collector is not particularly limited, and may be formed of any material so long as it has a suitable conductivity without causing chemical changes in the fabricated battery. Examples of the cathode current collector include copper, stainless steel, aluminum, nickel, titanium, sintered carbon, copper or stainless steel that is surface-treated with carbon, nickel, titanium or silver, and aluminum-cadmium alloys. In addition, the cathode current collector may be processed to have fine irregularities on surfaces thereof so as to enhance adhesive strength of the cathode current collector to the cathode active material, and may be used in any of various forms including films, sheets, foils, nets, porous structures, foams, and non-woven fabrics. The cathode active material layer includes a cathode active material, a binder, and a conducting agent.

(32) The anode active layer includes an anode active material, a binder, and optionally a conducting agent. The anode active material is not particularly limited, and may be selected from any anode active materials used in the art. Non-limiting examples of the anode active material are lithium

metal, a lithium metal alloy, a transition metal oxide, a doped or undoped lithium material, and a material that allows reversible intercalation and deintercalation of lithium ions, which may be used as a mixture or in combination of at least two thereof. The lithium metal alloy may be an alloy of lithium with a metal selected from sodium (Na), potassium (K), rubidium (Rb), cesium (Cs), francium (Fr), beryllium (Be), magnesium (Mg), calcium (Ca), strontium (Sr), silicon (Si), antimony (Sb), lead (Pb), indium (In), zinc (Zn), barium (Ba), radium (Ra), germanium (Ge), aluminum (Al), and tin (Sn). Non-limiting examples of the transition metal oxide are tungsten oxide, molybdenum oxide, titanium oxide, lithium titanium oxide, vanadium oxide, and lithium vanadium oxide. Examples of the material that allows doping or undoping of lithium therein are Si, Sn, Al, Ge, Pb, Bi, Sb, and a Si—Y alloy (where Y is an alkali metal, an alkali earth metal, a Group 11 element, a Group 12 element, a Group 13 element, a Group 14 element, a Group 15 element, a Group 16 element, a transition metal, a rare earth element, and a combination thereof, except for Sn. For example, Y may be magnesium (Mg), calcium (Ca), strontium (Sr), barium (Ba), radium (Ra), scandium (Sc), yttrium (Y), titanium (Ti), zirconium (Zr), hafnium (Hf), vanadium (V), niobium (Nb), tantalum (Ta), chromium (Cr), molybdenum (Mo), tungsten (W), seaborgium (Sg), technetium (Tc), rhenium (Re), iron (Fe), lead (Pb), ruthenium (Ru), osmium (Os), rhodium (Rh), iridium (Ir), palladium (Pd), platinum (Pt), copper (Cu), silver (Ag), gold (Au), zinc (Zn), cadmium (Cd), boron (B), aluminum (Al), gallium (Ga), tin (Sn), indium (In), titanium (Ti), germanium (Ge), phosphorus (P), arsenic (As), antimony (Sb), bismuth (Bi), sulfur (S), selenium (Se), tellurium (Te), or combinations thereof.

(33) The material that allows reversible intercalation and deintercalation of lithium ions may be any carbonaceous anode active material that is commonly used in a lithium battery. Examples of such carbonaceous materials are crystalline carbon, amorphous carbon, or mixtures thereof. Non-limiting examples of the crystalline carbon are natural graphite, artificial graphite, expanded graphite, graphene, fullerene soot, carbon nanotubes, and carbon fiber. Non-limiting examples of the amorphous carbon are soft carbon (carbon sintered at low temperatures), hard carbon, mesophase pitch carbides, and sintered corks. The carbonaceous anode active material may be in, for example, spherical, planar, fibrous, tubular, or powder form.

(34) The binder strongly binds anode active material particles together and to the anode current collector. Non-limiting examples of the binder are polyvinyl alcohol, carboxymethyl cellulose, hydroxypropyl cellulose, diacetyl cellulose, polyvinyl chloride, carboxylated polyvinyl chloride, polyvinyl fluoride, a polymer including ethylene oxide, polyvinylpyrrolidone, polyurethane, polytetrafluoroethylene, polyvinylidene fluoride, polyethylene, polypropylene, styrene-butadiene rubber (“SBR”), acrylated SBR, epoxy resin, and nylon.

(35) The conducting agent is used to provide conductivity to the anode. Any electron conducting material that does not induce chemical change in batteries may be used. Examples of the conducting agent are carbonaceous materials, such as natural graphite, artificial graphite, carbon black, acetylene black, ketjen black, carbon fibers, and the like; metal-based materials, such as copper (Cu), nickel (Ni), aluminum (Al), silver (Ag), and the like, in powder or fiber form; and conductive materials, including conductive polymers, such as a polyphenylene derivative, and mixtures thereof.

(36) The amounts of the anode active material, the binder, and the conducting agent may be equivalent to those commonly used in lithium batteries. For example, a weight ratio of the anode active material to a mixture of the conducting agent and the binder may be from about 98:2 to about 92:8. A mixing ratio of the conducting agent to the binder may be, but not limited to, from about 1:1.5 to about 1:3. The anode **112** and the cathode **114** may be each manufactured by mixing an active material, a conducting agent, and a binder in a solvent to prepare an active material composition, and coating the active material composition on a current collector. Any method of manufacturing such electrodes which is known to one of ordinary skill in the art may be used. Thus, a detailed description thereof will not be provided herein. Non-limiting examples of the

solvent are N-methylpyrrolidone (“NMP”), acetone, and water.

(37) A separator may be disposed between the cathode and the anode, according to the type of the lithium secondary battery. The separator helps maintain electrical isolation between the cathode and the anode. The separator may be any separator that is commonly used for lithium batteries. In particular, the separator may have low resistance to migration of ions in an electrolyte and have high electrolyte-retaining ability. Examples of the separator are glass fiber, polyester, Teflon, polyethylene, polypropylene, polyvinylidene fluoride (“PVDF”), polytetrafluoroethylene (“PTFE”), and a combination thereof, each of which may be a nonwoven fabric or a woven fabric. The separator may be a single layer or a multi-layer. Examples of the separator are a polyethylene/polypropylene double-layer separator, polyethylene/polypropylene/polyethylene triple-layer separator, and a polypropylene/polyethylene/polypropylene triple-layer separator. The separator may have a pore diameter of about 0.01 to about 10 μm and a thickness of about 3 to about 100 μm . The electrolyte may be injected between the cathode **114** and the anode **112** with the separator **113** therebetween.

(38) A separator may include fibers, particles, web, porous sheets, or other forms of material configured to reduce the risk of physical contact and/or short circuit between the electrodes. The separator may be a unitary element, or may include a plurality of discrete spacer elements such as particles or fibers. In some examples, the electrolyte layer may include a separator infused with an electrolyte solution. In some examples such as a polymer electrolyte, the separator may be omitted.

(39) See: U.S. Pat. Nos. 8,318,353; 7,901,810; 7,790,319; 7,771,628; 7,759,008; 7,041,239; 6,872,492; 4,355,162; 4,376,709; 4,376,709; 4,459,406; 5,849,434; 6,844,047; 6,872,492; 6,872,492; 6,888,663; 6,958,860; 7,041,239; 7,338,734; 7,383,083; 7,759,008; 7,759,008; 7,771,628; 7,790,319; 7,901,810; 7,939,201; 7,955,733; 7,960,058; 7,964,308; 7,972,728; 7,988,746; 7,998,617; 8,003,250; 8,057,936; 8,067,117; 8,088,512; 8,148,009; 8,148,013; 8,158,090; 8,168,326; 8,206,468; 8,206,469; 8,218,351; 8,241,789; 8,277,975; 8,282,691; 8,318,353; 8,323,831; 8,323,832; 8,361,661; 8,420,258; 8,435,678; 8,445,135; 8,449,980; 8,481,208; 8,492,031; 8,580,430; 8,586,238; 8,617,430; 8,617,745; 8,652,683; 8,658,310; 8,709,647; 8,722,227; 8,722,227; 8,722,242; 8,734,539; 8,785,043; 8,821,763; 8,852,807; 8,932,762; 8,936,871; 8,951,676; 8,951,680; 8,993,171; 8,999,571; 9,065,093; 9,077,032; 9,293,790; 9,299,966; 9,305,716; 9,349,544; 9,362,562; 20020192553; 20020192553; 20020192553; 20040005265; 20040016632; 20040016632; 20040048157; 20040066556; 20040197654; 20040197654; 20040262571; 20040262571; 20050238961; 20050238961; 20050238961; 20060110580; 20060194112; 20060194112; 20060292444; 20070031732; 20070072034; 20070072034; 20070141468; 20070141468; 20070166617; 20070190418; 20070248520; 20070292747; 20070298317; 20080261113; 20080261113; 20090214944; 20090220838; 20090220838; 20090246636; 20090311597; 20100075225; 20100078591; 20100233545; 20100266899; 20100266899; 20100303701; 20110052473; 20110052986; 20110052986; 20110052986; 20110052995; 20110159381; 20110195306; 20110200848; 20110200848; 20110229765; 20110229765; 20110274948; 20120140378; 20120164499; 20120164499; 20120214071; 20120219856; 20120219859; 20120237828; 20120270109; 20120315538; 20130034780; 20130052492; 20130059211; 20130078523; 20130078524; 20130084474; 20130084474; 20130084474; 20130115521; 20130252112; 20130252114; 20130302697; 20130344367; 20130344367; 20140099557; 20140220450; 20140242445; 20140302403; 20140370388; 20150111105; 20150132650; 20150236342; 20150236349; 20150263382; 20150303474; 20150303474; 20150311565; 20150364753; 20160028114; 20160096334; 20160111720; and 20160164152, each of which is expressly incorporated herein by reference in its entirety.

(40) See also: Berrah, Fadila, et al. “The vanadium monophosphates AVOPo.sub.4: Synthesis of a second form β -KVOPo.sub.4 and structural relationships in the series.” *Solid state sciences* 3.4 (2001): 477-482. Zima, Vitězslav, et al. “Ion-exchange properties of alkali-metal redox-intercalated

vanadyl phosphate.” *Journal of Solid State Chemistry* 163.1 (2002): 281-285. Lii, Kwang-Hwa, and Wei-Chuan Liu. “RbVOPO₄ and CsVOPO₄, Two Vanadyl (IV) Orthophosphates with an Intersecting Tunnel Structure and Discrete VO₅ Pyramids.” *Journal of Solid State Chemistry* 103.1 (1993): 38-44. Yakubovich, O. V., O. V. Karimova, and O. K. Mel'nikov. “The mixed anionic framework in the structure of Na₂{MnF [PO₄]}.” *Acta Crystallographica Section C: Crystal Structure Communications* 53.4 (1997): 395-397. Schindler, M., F. C. Hawthorne, and W. H. Baur. “Crystal chemical aspects of vanadium: polyhedral geometries, characteristic bond valences, and polymerization of (VO)_n polyhedra.” *Chemistry of Materials* 12.5 (2000): 1248-1259. Panin, Rodion V., et al. “Crystal Structure, Polymorphism, and Properties of the New Vanadyl Phosphate Na₄VO(PO₄)₂.” *Chemistry of materials* 16.6 (2004): 1048-1055. Belkhir, Sabrina, Djillali Mezaoui, and Thierry Roisnel. “The Structure Determination of a New Mixed Mono-Arsenate K₂V₂O₂(AsO₄)₂.” *3ème Conférence Internationale sur le Soudage, le CND et l'Industrie des Matériaux et Alliages (IC-WNDT-MI'12)*. Centre de Recherche Scientifique et Technique en Soudage et Contrôle (CSC), 2012. Glasser, Leslie, and C. Richard A. Catlow. “Modelling phase changes in the potassium titanyl phosphate system.” *Journal of Materials Chemistry* 7.12 (1997): 2537-2542. Fedotov, Stanislav S., et al. “AVPO₄F (A=Li, K): A 4 V Cathode Material for High-Power Rechargeable Batteries.” *Chemistry of Materials* 28.2 (2016): 411-415. Belkhir, Sabrina, Djillali Mezaoui, and Thierry Roisnel. “K₂V₂O₂(AsO₄)₂.” *Acta Crystallographica Section E: Structure Reports Online* 68.7 (2012): i54-i54. Yakubovich, O. V., V. V. Kireev, and O. K. Mel'nikov. “Refinement of crystal structure of a Ge-analogue of natisite Na₂{TiGeO₄} and prediction of new phases with anionic {MTOs} radicals.” *Crystallography Reports* 45.4 (2000): 578-584. Boudin, S., et al. “Review on vanadium phosphates with mono and divalent metallic cations: syntheses, structural relationships and classification, properties.” *International Journal of Inorganic Materials* 2.6 (2000): 561-579.

(41) Each reference cited herein is expressly incorporated herein by reference in its entirety.

SUMMARY OF THE INVENTION

(42) The present technology provides a vanadyl phosphates ε-VOPO₄ cathode which has achieved multi-electron storage as lithium-ion battery cathode. Vanadyl phosphates in general have low intrinsic conductivity. A high efficiency battery cathode should have low electrical resistance. To overcome the conductivity problem, the cathode material is preferably nanosized, and coated with particles of a low activation energy conductive material, such as graphene or carbon nanotubes. This cathode utilizes the two redox couples of vanadium cation (i.e. V⁵⁺/V⁴⁺, V⁴⁺/V³⁺) to permit more than one lithium ion to be stored in the unit structure per vanadium ion. The involvement of the multiple redox processes of vanadium is reflected by the well separated high voltage plateau region at ~3.8 V and low voltage plateau region at ~2 V.

(43) The two-electron redox property of vanadium results in a theoretical capacity of 305 mAh/g. In practical, maximum discharge capacity of over 300 mAh/g was obtained within the voltage region of 1.3-4.5 V vs. Li/Li⁺, which is over 90% of the theoretical value.

(44) The electrode material is not limited to use in batteries, or as a cathode, or for use in lithium-ion electrolyte systems.

(45) In addition to ε-VOPO₄ material, the cathode may further contain any cathode material suitable for lithium-ion insertion and release. Suitable auxiliary materials may include phosphate based materials such as FePO₄, VPO₄F, V₂(PO₄)₂F₃, FePO₄F, and V₂(PO₄)₃; oxides such as CoO₂, orthorhombic MnO₂, layered iron oxides FeO₂, chromium oxide CrO₂, layered Ni_{0.5}Mn_{0.5}O₂, and V₆O₁₅ nanorods; layer sulfides such as TiS₂; perovskite transition metal fluorides, or a mixture thereof.

(46) The epsilon polymorph of vanadyl phosphate, ε-VOPO₄, made from the hydrothermally or more generally, solvothermally synthesized H₂VOPO₄, is a cathode material for

lithium-ion batteries that has been optimized to reversibly intercalate two Li-ions to reach the full theoretical capacity at least 50 cycles with a coulombic efficiency of 98%. This material adopts a stable 3D tunnel structure and can extract two Li-ions per vanadium ion, giving a theoretical capacity of 305 mAh/g, with an upper charge/discharge plateau at around 4.0 V, and one lower at around 2.5 V. As ϵ -VOPO.sub.4 is capable of reversibly intercalating more than one lithium ion into the structure, it stores and delivers more energy than current batteries in the market. Compared to LiFePO.sub.4, ϵ -VOPO.sub.4 has a higher electronic conductivity and higher energy density with the insertion of one Li-ion, 159 mAh/g at 4.0 V vs 170 mAh/g at 3.45 V. Overall, ϵ -VOPO.sub.4 makes a great candidate for next generation of high energy Li-ion batteries. The nano-sized ϵ -VOPO.sub.4 particles demonstrate enhanced electrochemistry and cyclability for potential applications in lithium-ion batteries.

(47) Phosphate based materials have been considered as excellent cathode candidates because of their high stability and low cost. However, most phosphate cathodes show poor electronic conductivity and as a result, full capacity of the cathode can't be achieved in the traditional charge/discharge processes.

(48) One approach to obtain a cathode of high capacity is to employ a transition metal capable of multiple electron transfer, and thus able to assume more than one lithium. Vanadium is well-known to be capable of transfer of two electrons, such as from the +5 to +3 oxidation state.

(49) Vanadyl phosphate (VOPO.sub.4) is a material combining the merits of vanadium and of phosphate and theoretically has the possibility to show high capacity as well as good stability as a cathode active material for a sodium battery. The vanadyl phosphates with formula of AVOPO.sub.4 (A=alkali metal) form a class of materials which can serve as a multi-electron cathode. These cathodes can utilize the V.sup.3+-V.sup.4+-V.sup.5+ redox couples, during which two ions can be reversible stored in the structure instead of one. Therefore, this class of cathode materials is expected to exhibit much higher energy density than the traditional one-electron cathodes. By far, the two-electron behavior only has been observed in Li ion system (i.e. two-Li) in some different phases of VOPO.sub.4 and LiVOPO.sub.4, within a voltage window covering the V.sup.3+.fwdarw.V.sup.5+ transition, which exhibits enhanced practical energy densities.

(50) Vanadium phosphate materials have been described as cathode materials.

(51) It is an object of this invention to a high energy density cathode active material for use in a lithium battery, using ϵ -VOPO.sub.4 as an active material, and which comprises a conductivity enhancer comprising graphene or carbon nanotubes.

(52) The cathode containing any of the above-listed materials may be mixed with other electrically conductive materials and binders. Examples of electrically conductive materials include carbon black and vapor ground carbon fibers. Examples of binders include polyvinylidene fluoride (PVDF), sodium alginate, and sodium carboxymethyl cellulose.

(53) The cathode active material may be mixed with binders recognized by one of skill in the art as suitable for lithium-ion batteries. For example, suitable binders may include PVDF, polytetrafluoroethylene (PTFE), styrene butadiene rubber (SBR), and polyimide.

(54) According to the present invention, the conductive additive is graphene or carbon nanotubes. The amount of conductive additive may be 1-10% by weight, and preferably about 5%.

(55) The cathode active material may be present as a sheet, ribbon, particles, or other physical form. An electrode containing the cathode active material may be supported by a current collector. A current collector may include a metal or other electrically conducting material. The current collector may be formed of carbon, carbon paper, carbon cloth or a metal or noble metal mesh or foil.

(56) U.S. Pat. No. 9,722,247 (Zhang et al.) discusses F-Vanadyl phosphates as high energy density cathode materials for rechargeable sodium batteries.

(57) For example, U.S. Pat. No. 6,872,492 (Barker et al.) describes sodium ion batteries based on cathode materials of the general formula: A.sub.aM.sub.b(XY.sub.4).sub.cZ.sub.d. Example 4b

describes synthesis of VOPO.sub.4xH.sub.2O and Examples 4c and 4d describe synthesis of NaVOPO.sub.4. Charge and discharge of a cell containing a cathode of the NaVOPO.sub.4 and a negative electrode of lithium metal is described. Sodium ion cells prepared are based on a carbon composite negative electrode and NaVOPO.sub.4F as the positive electrode active material.

(58) U.S. 2013/0034780 (Muldoon et al.) describes a magnesium battery and lists VOPO.sub.4 as a suitable positive electrode active material.

(59) U.S. 2004/0048157 (Neudecker et al.) describes a lithium solid state thin film battery containing a lithiated vanadium oxide film as an anode and as one possible cathode material, LiVOPO.sub.4.

(60) U.S. 2013/0260228 (Sano et al.) describes a lithium secondary battery having as a positive electrode material, a compound of the formula: Lia(M).sub.b(PO.sub.4).sub.cF.sub.d. LiVOPO.sub.4 is described in a preferred embodiment.

(61) U.S. 2013/0115521 (Doe et al.) describes a magnesium secondary battery wherein the current collectors are coated with a thin protective coating. VOPO.sub.4 is listed as a positive electrode active material.

(62) U.S. 2012/0302697 (Wang et al.) describes a magnesium cell having a carbon or other graphitic material as a cathode active material. VOPO.sub.4 is included in a list of other cathode active materials.

(63) Kerr, T. A., J. Gaubicher, and L. F. Nazar. "Highly Reversible Li Insertion at 4 V in ϵ -VOPO.sub.4/ α -LiVOPO.sub.4 Cathodes." *Electrochemical and Solid-State Letters* 3, no. 10 (2000): 460-462 discusses a ϵ -VOPO.sub.4/ α -LiVOPO.sub.4 cathode for a lithium battery having a capacity of 100 mA/g.

(64) Quackenbush, Nicholas F., Linda Wangoh, Bohua Wen, Ruibo Zhang, Youngmin Chung, Natasha Chernova, Zehua Chen et al. "Interfacial Effects of Electrochemical Lithiation of Epsilon-VOPO4 and Evolution of the Electronic Structure." In *Meeting Abstracts*, no. 6, pp. 491-491. The Electrochemical Society, 2015 discusses the use of ϵ -VOPO.sub.4 as a cathode material in lithium batteries, which achieves dual cation exchange only on particle surfaces, and not in the bulk material.

(65) Seven distinct VOPO.sub.4 structures or phases are known. All of the reported structures contain VO6 octahedra sharing vertices with PO.sub.4 tetrahedra. The oxygen polyhedron of vanadium is irregular so that it is often considered as a VO.sub.5 square pyramid with a very short apical vanadyl bond (V=O) and a much more remote sixth oxygen atom (V . . . O). These seven phases can be distinguished as:

(66) α .sub.I: has a lamellar structure with alternating antiparallel V=O bonds pointing inside the layers.

(67) α .sub.II: also has a lamellar structure with antiparallel V=O bonds pointing outside the layers.

(68) γ : is an intermediate form between α .sub.I and α .sub.II with half parallel V=O bonds pointing inside, half outside the layers.

(69) δ : has antiparallel V=O bonds pointing half inside, half outside the layers. The vanadyl chains point to different directions in the unit cell.

(70) ω : shows disordered vanadyl chains in the [1 0 0] and [0 1 0] directions of the tetragonal cell.

(71) β : All vanadyl chains are parallel and tilted to form zigzag O=V . . . O=V chains.

(72) ϵ : The structure is a distorted form of β -phase and differs in terms of tilted O=V . . . O angle.

(73) It is therefore an object to provide an intercalation-type electrode composition for a lithium ion battery cathode, having a capacity of at least 275 mAh/g, and may have a capacity of at least 280 mAh/g, at least 290 mAh/g, at least 300 mAh/g, or at least 305 mAh/g, for example. The intercalation electrode composition may comprise a transition metal having a two-electron redox property, having a discharge capacity of at least 75% of theoretical value, at least 80% of theoretical value, at least 85% of theoretical value, or at least 90% of theoretical value.

(74) The lithium ion battery cathode may comprise a transition metal, which undergoes a change in

oxidation state of at least two between a charged and discharged state.

(75) The intercalation electrode composition preferably comprises VOPO.sub.4, most preferably in the epsilon form, i.e., ϵ -VOPO.sub.4. The intercalation electrode composition preferably has a conductivity enhancer comprising graphene or carbon nanotubes. The intercalation electrode composition may comprise ϵ -VOPO.sub.4 and at least 2.5% by weight graphene, at least 3.0% by weight graphene, at least 3.5% by weight graphene, at least 4.0% by weight graphene, at least 5% by weight graphene, at least 6% by weight graphene, at least 7% by weight graphene, at least 8% by weight graphene, at least 9% by weight graphene, or at least 10% by weight graphene.

(76) The intercalation electrode composition may comprise, for example, at least 75% by weight ϵ -VOPO.sub.4, at least 5% by weight graphene nanoplatelets, and at least 5% by weight of a poly vinylidene fluoride (PVDF) binder. The intercalation electrode composition may comprise 85% by weight ϵ -VOPO.sub.4, at least 5% by weight graphene nanoplatelets, and 10% by weight binder. The intercalation electrode composition may comprise 75% by weight ϵ -VOPO.sub.4, 15% by weight graphene nano platelets, and 10% by weight of a poly vinylidene fluoride (PVDF) binder.

(77) The intercalation electrode composition may be provided as a cathode in a battery comprising a lithium-ion anode, an electrolyte adapted to operate at a battery potential of at least 4.5V, a separator, and a supporting lithium salt.

(78) The intercalation electrode composition may have a first state in which at least 80 mol % of a transition metal element is oxidized in a first oxidation state, and a second state in which at least 80 mol % of a transition metal element is oxidized in a second oxidation state, the first state and the second state differing by two, and the at least 80 mol % of the transition metal element in the first state is associated with two lithium ions per transition metal element ion.

(79) Another object provides a lithium-ion battery cathode composition, comprising ϵ -VOPO.sub.4, electrically conductive graphene in a ratio of at least 3% by weight of the ϵ -VOPO.sub.4, and a binder, on a current collector substrate.

(80) A further object provides an intercalation electrode composition for a lithium-ion cathode, having a dual lithium-ion exchange characteristic, having a capacity of about 125 mAh/g at a voltage exceeding 3.7 V, and a capacity of about 260 mAh/g at a voltage exceeding 2.0 V.

(81) A still further object provides an intercalation electrode composition for a lithium-ion cathode, having a dual lithium-ion exchange characteristic, having an energy capacity of at least 850 mWh/g. The energy capacity may be at least 860 mWh/g, at least 870 mWh/g, an energy capacity of at least 880 mWh/g, an energy capacity of at least 890 mWh/g, or an energy capacity of at least 900 mWh/g.

(82) Another object provides an electrode composition for a lithium-ion cathode, comprising ϵ -VOPO.sub.4 having a theoretical capacity of 305 mA/g and an observed capacity of at least 275 mAh/g. The observed capacity may be at least 280 mAh/g, at least 285 mAh/g, at least 290 mAh/g, at least 295 mAh/g, or at least 300 mAh/g.

(83) A further object provides an electrode, comprising a ϵ -VOPO.sub.4 lithium ion exchange active material, graphene nanoplatelets, and a binder, having an energy density of 900 mWh/g.

(84) A still further object provides an electrode, comprising a ϵ -VOPO.sub.4 lithium ion exchange active material, having a current-voltage profile which displays peaks at about 2.1 V, 2.25 V, 2.5 V, and 3.9 V. representing a dual-lithium-ion exchange per vanadium ion of at least 90%.

(85) An object also provides a method of making a lithium-ion battery cathode, comprising: hydrothermally or solvothermally generating ϵ -VOPO.sub.4; mixing the ϵ -VOPO.sub.4, with graphene nanoplatelets and a binder to form a mixture; and coating a current collector with a slurry of the mixture. The graphene nanoplatelets may have a surface area of 750 m²/g, for example. The binder may be polyvinylidene fluoride. The weight ratio of ϵ -VOPO.sub.4, graphene nanoplatelets, and polyvinylidene fluoride may be 75:15:10.

(86) The invention comprises the cathode material, an electrode formed of the cathode material,

and electrochemical devices, e.g., a secondary battery, formed using the electrode.
(87) Other objects will be apparent from a review hereof.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) These and/or other aspects will become apparent and more readily appreciated from the following description of the embodiments, taken in conjunction with the accompanying drawings in which:
- (2) FIG. 1A shows morphological and structure characterization of ϵ -VOPO.sub.4 SEM image.
- (3) FIG. 1B shows an XRD pattern with Rietveld refinement of the as-synthesized ϵ -VOPO₄.
- (4) FIG. 2 shows TEM images of ϵ -VOPO.sub.4 hand ground with graphene nanoplatelets for electrode preparation.
- (5) FIG. 3A shows galvanostatic charge-discharge curves of ϵ -VOPO.sub.4 from 1.6 to 4.5 V at C/50.
- (6) FIG. 3B shows cycle performance of ϵ -VOPO.sub.4 from 1.6 to 4.5 V at C/50, 1C=2 Li.
- (7) FIG. 4 shows a CV curve profile of ϵ -VOPO.sub.4 at a scan rate of 0.02 mV/s.
- (8) FIG. 5A shows galvanostatic charge-discharge curves of ϵ -VOPO.sub.4 from 1.6 to 4.5 V at C/20.
- (9) FIG. 5B shows cycle performance of ϵ -VOPO₄ from 1.6 to 4.5 V at C/20, 1C=2Li.
- (10) FIG. 6A shows galvanostatic charge-discharge curves of ϵ -VOPO.sub.4 at the low voltage region, from 1.6 to 3.0 V.
- (11) FIG. 6B shows cycle performance in the low voltage region, 1.6 to 3.0 V, of ϵ -VOPO.sub.4 at C/50, 1C=2Li.
- (12) FIG. 7A shows cycling curves of ϵ -VOPO.sub.4 in the low voltage region, from 1.6-3.0 V, at different rates.
- (13) FIG. 7B shows rate test capacities of ϵ -VOPO.sub.4 in the low voltage region, from 1.6 to 3.0 V.
- (14) FIG. 8A shows galvanostatic charge-discharge curves of ϵ -VOPO.sub.4 at the high voltage region, from 3.0 to 4.5 V.
- (15) FIG. 8B shows cycle performance in the high voltage region, 3.0-4.5 V, of ϵ -VOPO.sub.4 at C/50, 1C=2Li.
- (16) FIG. 9A shows Cycle curves of ϵ -VOPO.sub.4 at high voltage region, from 3.0 to 4.5 V, at different current rates.
- (17) FIG. 9B shows rate test capacities of ϵ -VOPO.sub.4 in the low voltage region, from 3.0 to 4.5 V.
- (18) FIG. 10 is an exploded perspective view of a lithium secondary battery according to an embodiment.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

- (19) Reference will now be made in detail to embodiments, examples of which are illustrated in the accompanying drawings, wherein like reference numerals refer to the like elements throughout. In this regard, the present embodiments may have different forms and should not be construed as being limited to the descriptions set forth herein. Accordingly, the embodiments are merely described below, by referring to the figures, to explain aspects of the present description. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. Expressions such as “at least one of,” when preceding a list of elements, modify the entire list of elements and do not modify the individual elements of the list.
- (20) It will be understood that when an element is referred to as being “on” another element, it can be directly in contact with the other element or intervening elements may be present therebetween.

In contrast, when an element is referred to as being “directly on” another element, there are no intervening elements present.

(21) It will be understood that, although the terms first, second, third etc. may be used herein to describe various elements, components, regions, layers, and/or sections, these elements, components, regions, layers, and/or sections should not be limited by these terms. These terms are only used to distinguish one element, component, region, layer, or section from another element, component, region, layer, or section. Thus, a first element, component, region, layer, or section discussed below could be termed a second element, component, region, layer, or section without departing from the teachings of the present embodiments.

(22) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. The term “or” means “and/or.” It will be further understood that the terms “comprises” and/or “comprising,” or “includes” and/or “including” when used in this specification, specify the presence of stated features, regions, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, regions, integers, steps, operations, elements, components, and/or groups thereof. Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this general inventive concept belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, are appropriate for use only if consistent with a meaning that is consistent with their meaning in the context of the relevant art and the present disclosure, and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein. Thus, such resources as “Merriam Webster” (any version) are secondary to field of science-appropriate technical dictionaries and encyclopedias.

(23) One or more embodiments will now be described in detail with reference to the following examples. However, these examples are not intended to limit the scope of the one or more embodiments.

(24) The basis of the lithium-ion battery (LIB) uses lithium-ions to travel across the electrolyte and intercalate into the anode upon charge and into the cathode upon discharge [1]. Because they outperform competing primary batteries like lead-acid, alkaline, etc., as well as other rechargeable batteries such as nickel-metal hydride, nickel cadmium, etc., it is no wonder how the LIB has revolutionized and expanded the mobile electronics industry since 1991 [2]. While the performance and functionality of smartphones and laptops continue to improve, the development of LIBs need to catch up to match in terms of power and life cycle to expand into large energy storage applications.

(25) Currently, the cathode material in the market is dominated by LiCoO_2 . While it has an extremely high theoretical capacity of 274 mAh/g with an operating voltage around 3.6 V, the structure tends to undergo irreversible changes when more than 50% of the Li-ions are removed [3]. Environmental safety and the cost became major concerns as well, thus, drove the search for alternatives where the scarce and toxic cobalt is substituted by other metals such as in $\text{LiNi}_{1/3}\text{Mn}_{1/3}\text{Co}_{1/3}\text{O}_2$, also known as NMC [4]. Extensive research for developing alternative cathode materials lead to study of metal phosphates, thus came the LiFePO_4 . This olivine material has garnered a great deal of attention that it has been commercialized for portable and stationary systems by A123, BAE Systems and in China. In fact, Hydro-Quebec, MIT and A123 further improved this material with high-power performance that can charge and discharge within minutes through nanosizing and carbon coating [5]. Through a collaborative effort from NECCES-I, a fundamental study on this phenomenon determined a metastable reaction mechanism of LiFePO_4 which is why it can cycle at high rates [6].

(26) Another way to leverage in the stability of phosphates and to increase energy storage is to incorporate a second electron. Hautier et al. plotted the mean voltage for each per metal redox

couple with respect to the capacity in a phosphate, giving way to consider two electron couples. Vanadium phosphate compounds were candidates within the acceptable voltage window [7]. The ϵ -VOPO.sub.4.Math. ϵ -LiVOPO.sub.4.Math. ϵ -Li.sub.2VOPO.sub.4 system has been regarded as one of the most promising and safe candidates to provide a two-electron reaction with a high theoretical capacity of 305 mAh/g and specific energy over 900 Wh/g [8, 15, 16]. This system has two redox potentials at useful potentials for storing energy, V.sup.3+.Math.V.sup.4+ at 4.0 V and V.sup.4+.Math.V.sup.5+ at 2.5 V. ϵ -VOPO.sub.4 was first synthesized by Lim et al. by heating monoclinic H.sub.2VOPO.sub.4 in oxygen, and Kerr et al. measured the electrochemical reversibility at the high voltage plateau at 4.0 V [9, 10]. Previously, the synthesis and characterization of ϵ -VOPO.sub.4 from two different phases of H.sub.2VOPO.sub.4 was reported, to discover that the electrochemical performance from the disordered tetragonal precursor was improved due to smaller particle size [11]. Fundamental studies were conducted on the structural evolution of ϵ -LiVOPO.sub.4 and two intermediate phases in the low-voltage regime identified using DFT calculations backed up with X-ray pair distribution function analysis and X-ray absorption near edge structure measurements [12]. The insertion of two Li-ions into ϵ -VOPO.sub.4 has been demonstrated, reaching the theoretical specific capacity of 305 mAh/g.

Example 1

(27) Synthesis: ϵ -VOPO.sub.4 was synthesized by calcining the monoclinic H.sub.2VOPO.sub.4 precursor, as reported by Song et al. [11] Stoichiometric amounts of VCl.sub.3 (Sigma-Aldrich, 97%), and P.sub.2O.sub.5 (Sigma-Aldrich, $\geq$ 98%) were dissolved in 190 proof ethanol (Pharmco-AAPER). The solution was placed in a 4748 Type 125 mL PTFE-lined reactor (Parr Instrument Co.) and the reaction was set to 180° C. for 72 hours. The solvothermal product was collected by centrifugation and heated at 550° C. in flowing oxygen for 3 hours.

(28) Materials Characterization: For XRD measurements, a Bruker D8 Advanced X-ray diffractometer equipped with Cu K α source, λ =1.54178 Å. The intensities were recorded within the 2 θ range from 10° to 80° with 2 θ steps of 0.02° from powder samples. The unit cell parameters were obtained by Rietveld refinement with the TOPAS program. Scanning electron microscopy (SEM) measurements were performed with a Zeiss Supra-55 VP field emission scanning electron, using both the secondary electron and InLens modes to determine the morphology and particle size, at an acceleration voltage of 10 kV. Transmission electron microscopy (TEM) imaging was performed using the FEI Titan 80-300 microscope with a field emission gun (FEG) and an image aberration corrector, operated at an acceleration voltage of 300 kV. The pristine material was dispersed on a copper grid coated with a lacey carbon film for high-resolution transmission electron microscopy (HRTEM) observation.

(29) Electrochemistry: The electrodes were prepared by mixing the active material, ϵ -VOPO.sub.4, with graphene nanoplatelets (surface area 750 m.sup.2/g, XG Sciences) as a carbon additive and polyvinylidene fluoride (PVDF, Aldrich) binder in a weight ratio of 75:15:10. The slurry was created by adding 1-methyl-2-pyrrolidinone (NMP, Aldrich) which was then laminated onto an aluminum foil 144 current collector and vacuum-dried overnight before use. The dried electrodes, of area 1.2 cm.sup.2, contained 8-10 mg of active material and were assembled in 2325-type coin cells in a He-filled glovebox with a pure lithium chip (thickness 0.38 mm, Aldrich) as the counter and reference electrode. The electrolyte used was lithium hexafluorophosphate (1 M LiPF.sub.6) dissolved in ethylene carbonate (EC) and dimethyl carbonate (DMC) in 1:1 volume ratio with Celgard 2400 (Hoechst Celanese) as the separator. The electrochemical properties were investigated using the Bio-Logic VMP multichannel potentiostat. The cells were cycled galvanostatically in the high voltage region (3.0-4.5 V), low voltage region (1.6-3.0 V) and the whole voltage (1.6-4.5 V) window at C/50, where 1 C=2 Li or 305 mAh/g per gram of Li.sub.2VOPO.sub.4. Cells were also cycled at C/20 over the whole voltage range.

(30) Hydrothermal or solvothermal synthesis has many unique advantages because it offers good control over the sample's purity and crystallinity, easy to scale up and low cost. This method can

keep the overall particle size small and size distribution narrow which are vital features for good cathodic electrochemical performance. FIG. 1A shows the as-synthesized ϵ -VOPO.sub.4 powder as nano-sized primary particles, ~100-200 nm, that are cuboid in shape. This material matched well with earlier reported results from Chen et al., where ϵ -VOPO.sub.4 synthesized from monoclinic H.sub.2VOPO.sub.4 are made up of single crystals up to 200 nm [13]. Achieving small primary particles is important because it can improve the rate property for Li intercalation. Azmi et al. reported that smaller LiVOPO.sub.4 particle size results in easier lithium-ion diffusion with enhanced columbic efficiency by improving the capacity of lithium deintercalation upon discharge and the decreasing the lithium intercalation potential upon charge [14]. It is also observed in FIG. 1A, that the nano-sized ϵ -VOPO.sub.4 primary particles do not agglomerate nor form into secondary particles, providing a good surface area for the graphene or carbon nanotube additive to wrap around and assist in electron migration during the charge/discharge process.

(31) The lack of primary particle agglomeration is attributed to the choice of solvent used for synthesis. By using 190 proof ethanol, ϵ -VOPO.sub.4 results in loose particle morphology whereas 200 proof ethanol results in the formation of 2 μ m balls as secondary particles. By using this solvothermal synthesis route, the precursor was successfully synthesized and calcined to produce pure crystalline P-VOPO.sub.4, as seen in FIG. 1B, resulting in sharp and narrow peaks in the x-ray diffraction pattern. The observed pattern matched very well to the calculated pattern with no impurities or other vanadyl phosphate phases, resulting in a low R.sub.wp value of 4.55%.

(32) Normally, ball-mill treatment is necessary to break up any agglomeration and secondary particles and to reduce the particle size for good electrochemical performance. However, since the as-synthesized ϵ -VOPO.sub.4 is of nanometer size, there is no need to use this application, which helps preserve the crystal structure for better reversible intercalation chemistry. FIG. 2 shows HRTEM images of 75 wt. % ϵ -VOPO.sub.4 that was hand milled with 15 wt. % graphene nanoplatelets in a mortar and pestle before adding 10 wt. % PDVF and NMP for electrode preparation. FIGS. 2A and 2B show graphene nanoplatelets forming a conductive network between every single ϵ -VOPO.sub.4 primary particle. Upon closer inspection at FIG. 2B, HRTEM shows that the graphene nanoplatelets coated on the ϵ -VOPO4 particle is around 10 nm thick.

(33) Electrochemistry of ϵ -VOPO.sub.4:

(34) FIGS. 3A and 3B show ϵ -VOPO.sub.4 cycled in the whole voltage window from 1.6 V to 4.5 V at C/50, capable of achieving a high discharge capacity of 305 mAh/g for at least 50 cycles. FIG. 2A displays the desired characteristic plateaus at ~4.0 V at the high voltage region and at ~2.5, 2.25, 2.0 V at the low voltage region. The drop from the high voltage region to the low voltage region is a step-like curve and the hysteresis gap between the charge and discharge curve is very small. The high voltage region has a long plateau which extends the capacity to ~150 mAh/g, equivalent to ~1 Li. This corresponds to the redox potential of V.sup.3+ and V.sup.4+ where ϵ -VOPO.sub.4 becomes ϵ -LiVOPO.sub.4. The low voltage region has three plateaus at 2.5, 2.25 and 2.0 V which also extends the capacity to ~150 mAh/g, corresponding to the second intercalation of lithium where ϵ -LiVOPO.sub.4 becomes ϵ -Li.sub.2VOPO.sub.4. The plateaus at the low voltage region has maintained step-like curves even after 35 cycles, suggesting good kinetics and the changes in the local structure may be reversible for easy Li intercalation.

(35) Cyclic voltammetry (CV) curves was measured in the voltage window of 1.6 V to 4.5 V to understand the redox process of ϵ -VOPO.sub.4 is shown in FIG. 4. There are four reduction peaks at certain voltages that correspond to four oxidation peaks at similar voltages. Each peak represents the reversible reaction between ϵ -VOPO.sub.4 and ϵ -Li.sub.2VOPO.sub.4 that correspond to the voltage plateaus found upon galvanostatic charge and discharge cycling. Starting from the OCV point at 3.9 V, there is a single oxidation peak at 3.7 V that indicates electrochemical lithiation from ϵ -VOPO.sub.4 to ϵ -LiVOPO.sub.4. As the scan rate test moves to the low voltage region, there are three additional oxidation peaks. Each of the peaks signify the transition from ϵ -LiVOPO.sub.4 to ϵ -Li.sub.2VOPO.sub.4 with intermediate stages in between. ϵ -LiVOPO.sub.4 becomes ϵ -

Li.sub.1.5VOPO.sub.4 at ~2.5 V, then it converts to ϵ -Li.sub.1.75VOPO.sub.4 at ~2.25 V and finally becomes ϵ -Li.sub.2VOPO.sub.4 at ~2.0 V. Reduction peaks appear as the voltage continues to sweep from the low to high voltage domain, indicating that the V.sup.5+ oxidation state of ϵ -VOPO.sub.4 was recovered from V.sup.3+ of ϵ -Li.sub.2VOPO.sub.4. From ϵ -Li.sub.2VOPO.sub.4, it becomes ϵ -Li.sub.1.75VOPO.sub.4 at ~2.1 V, then ϵ -Li.sub.1.5VOPO.sub.4 at ~2.25 V and ϵ -LiVOPO.sub.4 at ~2.5 V. No further reaction takes place until ~4.25 V where ϵ -LiVOPO.sub.4 further reduces to become ϵ -VOPO.sub.4.

(36) FIGS. 5A and 5B show that even at a faster rate, ϵ -VOPO.sub.4 can still deliver a discharge capacity of ~305 mAh/g for up to 40 cycles at C/20. The long high voltage plateau extending past 100 mAh/g is preserved and each of the characteristic steps in the low voltage region are clearly sustained with no signs of diminishing for up to 30 cycles. In FIG. 4A, the drop from the high voltage region to the low voltage region evolved to a slope-like curve, which helps make up for the shorter high voltage plateau in the beginning but might indicate a little hysteresis. In subsequent cycles, the high voltage plateau slightly increases. The capacity of the 1.sup.st high voltage discharge plateau was ~125 mAh/g and by the 35.sup.th cycle, it increased to ~150 mAh/g which is equivalent to 1 Li. The low voltage region seems to show the opposite trend. As the high voltage plateau starts to increase in capacity, the low voltage steps start to decrease as well to maintain the overall discharge capacity at ~305 mAh/g.

(37) Cycling of ϵ -VOPO.sub.4 has been separated at the high voltage region and the low voltage region to study the stability of the electrochemical curve and capacity without the influence of each other. FIGS. 6A and 6B show ϵ -VOPO.sub.4 cycled in the low voltage region to study how long-term cycling affects the shape and length of these three distinct plateaus at 2.5 V, 2.25 V and 2.0 V which is in agreement with DFT calculations confirming the two intermediate phases at x=1.5 and 1.75 in the low-voltage regime [12]. The initial discharge curve in FIG. 6A is different because the cell was discharged from OCV first, delivering more than 300 mAh/g. Even after 30 cycles, FIG. 6A shows that each voltage step is clearly distinguished, delivering a reversible capacity of ~160 mAh/g, correlating to 1 Li. From then on, the cell was continuously charged and discharged in the low voltage window, from 1.6 V to 3.0 V. FIG. 6B shows that the low voltage steps maintained ~160 mAh/g for up to 30 cycles with no sign of decay at all, suggesting good kinetics at the low voltage region. The charge-discharge profiles with steps agree with many reports, giving evidence to the existence of intermediate phases of ϵ -Li.sub.xVOPO.sub.4 where x=1.5 and 1.75.

(38) A rate test in the low voltage region was performed to study how faster cycling can affect the plateaus at 2.5 V, 2.25 V and 2.0 V, as shown in FIGS. 7A and 7B. To start the rate test at the low voltage region, the cell was first discharged from OCV to 1.6 V at C/50, delivering a discharge capacity of over 300 mAh/g. From C/50 to C/5, the low voltage plateaus still maintained a discharge capacity of ~150 mAh/g with clearly defined step-like features, as shown in FIG. 7A. When the rate increased to 1C, the discharge capacity is still ~150 mAh/g but the plateaus are more slope-in shape at slightly lower voltages. As the cycling rate increases, the difference between the charge and discharge capacities decreases, as shown in FIG. 7B. From C/50, the charge capacity is 175 mAh/g while the discharge is ~150 mAh/g. When the rate increased to 1C, the charge and discharge capacities are ~150 mAh/g, thereby increasing the coulombic efficiency to ~100%. When cycled at the low voltage range, ϵ -VOPO.sub.4 can reversibly intercalate one full lithium ion at the low voltage region, even at faster cycling rates.

(39) FIGS. 8A and 8B show ϵ -VOPO.sub.4 cycled in the high voltage region, from 3.0 V to 4.5 V, to study how long-term cycling affects the shape and capacity. In this high voltage window, there is a plateau at ~4.0 V that coincides with the V.sup.3+/4+ redox where ϵ -VOPO.sub.4.Math.LiVOPO.sub.4. This high voltage plateau delivers a reversible capacity of ~140 mAh/g for up to 35 cycles which is close to 0.93 Li. This exceeds the previously reported results, where only 0.83 Li was inserted into ϵ -VOPO.sub.4 and 0.65 Li was inserted into ϵ -LiVOPO.sub.4. [11,12] After 30 cycles, the capacity slowly climbs to 150 mAh/g which corresponds to 1 Li which

agrees with the trend in the galvanostatic charge-discharge curves in FIG. 4. Even after 30 cycles, FIG. 8B shows that ϵ -VOPO.sub.4 delivers a reversible capacity of ~ 150 mAh/g, correlating to 1 Li. This plateau is step-like with no signs of fading after many cycles, suggesting easy reversible intercalation.

(40) FIGS. 9A and 9B show how different rates can affect the high voltage plateau of ϵ -VOPO.sub.4 at ~ 4.0 V. At C/50, the discharge capacity is around 130 mAh/g and the capacity decreases as the rate gets faster. By 1 C, the discharge capacity dropped to around 40 mAh/g. Despite the fast rate cycling of 1 C, the cell could deliver the high discharge capacity of 140 mAh/g after it was cycled back to C/50. This suggests that the structure was preserved, even at fast cycling, and can maintain a high discharge capacity when it was cycled back to C/50 from 1 C. It also seems that faster cycling leads to higher coulombic efficiency. It is evident that from C/50 to C, the coulombic efficiency increases. From C/25, some of the charge and discharge capacities are overlapping and by C/10, C/5 and C, the discharge capacities are practically the same as the charge. This means that at faster rates, it can de/intercalate lithium ion more efficiently.

CONCLUSIONS

(41) The optimized morphology and nano particle size of ϵ -VOPO.sub.4 is studied, as well as observations from transmission electron microscope to analyze the good carbon conductive network. By combining complementary characterization techniques of SEM, XRD and extensive electrochemical studies, the reversibility reaction of ϵ -VOPO.sub.4 as a cathode material for lithium-ion batteries is elucidated.

(42) Throughout this description all ranges described include all values and subranges therein, unless otherwise specified. Additionally, the indefinite article “a” or “an” carries the meaning of “one or more” throughout the description, unless otherwise specified. Where a numerical limit or range is stated, the endpoints are included. Also, all values and subranges within a numerical limit or range are specifically included as if explicitly written out.

(43) The above description is presented to enable a person skilled in the art to make and use the invention, and is provided in the context of a particular application and its requirements. Various modifications to the preferred embodiments will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other embodiments and applications without departing from the spirit and scope of the invention. Thus, this invention is not intended to be limited to the embodiments shown, but is to be accorded the widest scope consistent with the principles and features disclosed herein. In this regard, certain embodiments within the invention may not show every benefit of the invention, considered broadly.

REFERENCES

(44) M. S. Whittingham, Chem. Rev. 2014, 104, 4271-4301. M. Winter, R. J. Brodd, Chem. Reviews, 2004, 104, 4245-4270. B. C. Melot, J. M. Tarascon, Acc. Chem. Res., 2013, 46, 1226-1238. M. S. Whittingham, Chem. Rev. 2014, 114, 11414-11443. K. Zaghib, A. Mauger, F. Gendron, C. M. Julien, Chem. Mater., 2008, 20, 462-469. H. Liu, F. C. Strobridge, O. J. Borkiewicz, K. M. Wiaderek, K. W. Chapman, P. J. Chupas, C. P. Grey, Science, 2014, 344, 1252817. G. Hautier, A. Jain, S. P. Ong, B. Kang, C. Moore, R. Doe, G. Ceder, Chem. Mater. 2011, 23, 3495-3508. B. Wen, Q. Wang, Y. C. Lin, N. A. Chernova, K. Kharki, Y. Chung, F. Omenya, S. Sallis, L. F. J. Piper, S. P. Ong, M. S. Whittingham, Chem. Mater, 2016, 9, 3159-3170. S. C. Lim, J. T. Vaughey, W. T. A. Harrison, L. L. Dussack, A. J. Jacobson, J. W. Johnson, Solid State Ionics 1996, 84, 219-226. T. A. Kerr, Solid-State Lett. 1999, 3, 460. Song, Y.; Zavalij, P. Y.; Whittingham, M. S J. Electrochem. Soc. 2005, 152, A721. Y. C. Lin, B. Wen, K. M. Wiaderek, S. Sallis, H. Liu, S. H. Lapidus, O. J. Borkiewicz, N. F. Quackenbush, N. A. Chernova, K. Karki, F. Omenya, P. J. Chupas, L. F. J. Piper, M. S. Whittingham, K. W. Chapman, and S. P. Ong, Chem. Mater., 2016, 28, 1794-1805. Z. Chen, Q. Chen, L. Chen, R. Zhang, H. Zhou, N. A. Chernova, M. S. Whittingham, J. Electrochem. Soc. 2013, 160, A1777-A1780. B. M. Azmi, H. S. Munirah, T. Ishihara, Ionics, 2005, 11. M. Bianchini, J. M. Ateba-Mba, P. Dagault, E. Bogdan, D. Carlier, E. Suard, C. Masquelier, L. Croguennec, J.

Claims

1. A cathode composition, comprising ϵ -VOPO.sub.4 nanoparticles in an amount of at least 75% by weight, coated with graphene nanoplatelets in an amount of at least 3% by weight, wherein the graphene nanoplatelets form a conductive network between the ϵ -VOPO.sub.4 nanoparticles.
2. The cathode composition according to claim 1, wherein the ϵ -VOPO.sub.4 nanoparticles are solvothermally or hydrothermally generated.
3. The cathode composition according to claim 1, wherein the graphene nanoplatelets form a layer 10 nm thick on a surface of the ϵ -VOPO.sub.4 nanoparticles.
4. The cathode composition according to claim 1, wherein the ϵ -VOPO.sub.4 nanoparticles are between 100 nm and 200 nm in diameter.
5. The cathode composition according to claim 1, wherein the ϵ -VOPO.sub.4 nanoparticles are intercalated with lithium ions.
6. The cathode composition according to claim 1, wherein the ϵ -VOPO.sub.4 nanoparticles are adapted to intercalate dual lithium ions per vanadium, with an efficiency of at least 80%.
7. The cathode composition according to claim 1, wherein the ϵ -VOPO.sub.4 nanoparticles are adapted to intercalate dual lithium ions per vanadium, with an efficiency of at least 90%.
8. The cathode composition according to claim 1, wherein the graphene nanoplatelets have a surface area of 750 m² per gm.
9. The cathode composition according to claim 1, wherein the graphene nanoplatelets are coated on the ϵ -VOPO.sub.4 nanoparticles by milling.
10. The cathode composition according to claim 1, further comprising a binder in an amount of at least 2% by weight, adapted to bind the graphene nanoplatelet coated ϵ -VOPO.sub.4 nanoparticles together.
11. The cathode composition according to claim 10, wherein the binder comprises polyvinylidene fluoride.
12. A battery comprising: the cathode composition comprising ϵ -VOPO.sub.4 nanoparticles in an amount of at least 75% by weight, coated with graphene nanoplatelets in an amount of at least 3% by weight, wherein the graphene nanoplatelets form a conductive network between the ϵ -VOPO.sub.4 nanoparticles; an electrolyte with a lithium salt; a separator; and an anode.
13. The battery according to claim 12, wherein the anode comprises lithium metal.
14. The battery according to claim 12, having a first voltage plateau region at ~3.8 V and a second voltage plateau region at ~2 V.
15. The battery according to claim 12, having a power density of at least 850 mWh/g.
16. The battery according to claim 12, having an energy density of at least at least 275 mAh/g at a discharge rate of C/20.
17. A cathode for a rechargeable lithium ion battery, comprising: ϵ -VOPO.sub.4 nanoparticles; graphene nanoplatelets coating the ϵ -VOPO.sub.4 nanoparticles to form a conductive network; and a binder adapted to bind the graphene nanoplatelet coated ϵ -VOPO.sub.4 nanoparticles together and maintain the conductive network.
18. The cathode according to claim 17, further comprising a current collector on at least a surface of the cathode, adapted to communicate an electrical current with the conductive network.
19. The cathode according to claim 17, wherein the ϵ -VOPO.sub.4 nanoparticles are solvothermally or hydrothermally generated, the graphene nanoplatelets form a layer 10 nm thick on a surface of the ϵ -VOPO.sub.4 nanoparticles, and the ϵ -VOPO.sub.4 nanoparticles are between 100 nm and 200 nm in diameter.

20. The cathode according to claim 17, wherein at least a portion of the ϵ -VOPO.sub.4 nanoparticles are dual intercalated with lithium ions.
